ONE MISSING EDGE: THE EXACT $K_N - e$ CONE, STRICT SUPPORT, AND SHARP DEFECT LAWS

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ABSTRACT. Let $\mathcal{E}_{>0}(G)$ and $\mathcal{E}_{\geq 0}(G)$ denote the spectral regions of graph Laplacians with, respectively, positive and nonnegative weights on the allowed edges, and let $P(G)$ be the normalized volume of the latter region. Every noncomplete connected graph on $n \geq 3$ vertices satisfies the sharp pointwise obstruction

$$\lambda_n \geq \frac{n}{n-2}\lambda_2.$$

Equality forces a rigid block form and a matching complement. For every n, the obstruction is the exact relaxed cone of $K_n - e$:

$$\overline{\mathcal{E}_{>0}(K_n - e)} = \mathcal{E}_{\geq 0}(K_n - e) = \left\{\lambda_n \geq \frac{n}{n-2}\lambda_2\right\}.$$

An isospectral first-contact argument proves sufficiency; a codimension-two analysis of double contacts gives exact strict support for $n \geq 5$. At order four strict support fails only on $(t, t, 2t)$. Across all connected noncomplete graphs of a fixed order, the same half-cone is the exact union of strict spectral regions. We also obtain an exact positive-semidefinite symmetric doubly stochastic inverse-eigenvalue slice with off-diagonal support $K_n - e$.

At the next defect frontier, with M_2 the two-edge matching, every linear complement-orbitope obstruction for $K_n - M_2$ collapses to

$$\lambda_n + \frac{n-4}{n}\lambda_{n-1} \geq \lambda_2 + \lambda_3.$$

The two-level slice $\{0, a^{(n-3)}, b^{(2)}\}$ is relaxedly realizable exactly when $b/a \geq n/(n-2)$; the strict endpoint is classified as well. A spectral-pair insertion theorem also yields the exact one-spike classification. Moreover,

$$P(K_n - e) = 1 - \binom{(n-2)(n+1)/2}{n-2}^{-1}.$$

Together with the known paw and C_4 cones, the $K_4 - e$ result completes a boundary-linearity question of Fallat–Gupta–Lin. For every fixed nonempty graph H, the graph join satisfies $1 - P(H \vee K_k) = O_H(e^{-c_H k})$ for some $c_H > 0$, resolving an iterated-suspension question of Erchenko–Jakobson–Tsypin. Finally, one- and two-edge defects obey the sharp law $1 - P(G_n) = \exp\{-n \log n + O(n)\}$.

1. INTRODUCTION

For a finite simple graph G, the weighted Laplacian inverse eigenvalue problem asks which multisets can occur as the spectrum of a Laplacian whose positive edge weights have precisely the support G. Even when the graph is small, the answer is a semialgebraic region rather than a finite list. Every positive list of nonzero eigenvalues is realizable on the complete graph [6, Theorem 2.5]. The missing-edge inequality proved below shows that no other connected graph has this flexibility. The first missing edge is therefore the natural place at which a genuine obstruction must appear.

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Fallat, Gupta, and Lin solved $P_3 = K_3 - e$, determined the repeated-eigenvalue family $\{0, \lambda^{(2)}, \mu\}$ for $K_4 - e$, and asked whether the graph-dependent boundaries for the paw, C_4, and $K_4 - e$ are linear [6, Proposition 2.3, Theorem 3.15, Section 5, item 1]. Erchenko, Jakobson, and Tsypin subsequently determined the closed realizable regions for C_4 and the paw (called the kite in their paper), respectively [5, Theorems 4.1 and 6.1]. Catral, Fallat, Gupta, and Lin classified the matrices in $\mathcal{S}_L(K_n - e)$ that fail the strong spectral property for weighted Laplacians (SSPWL); their theorem supplies candidate boundary strata but does not determine which side is globally realizable [3, Theorem 4.2]. We determine both the exact closed region and the exact strict-support region for $K_n - e$ in every order. Together with the first two closed-cone descriptions just cited, this settles the remaining case of the boundary-linearity question. We also obtain dimension-uniform strict constructions and asymptotic theorems.

To the best of our knowledge, the full $K_n - e$ cone in arbitrary order, its exact strict-support locus, the resulting exact spectral-volume formula, and the fixed-core repeated-suspension convergence theorem have not previously appeared. This paper is entirely finite-dimensional. The genus-three comment in Remark 6.7 identifies only a possible leading tangent support; no graph-to-surface lifting or hyperbolic spectral prescription is asserted here.

Our first result is an obstruction that uses only the missing matrix entry. If G is a noncomplete connected graph on $n \geq 3$ vertices and

$$0 = \lambda_1 < \lambda_2 \leq \cdots \leq \lambda_n$$

is the spectrum of a strictly weighted Laplacian on G, then

$$\lambda_n \geq \frac{n}{n-2}\lambda_2. \tag{1}$$

The universal constant is sharp in every order, as witnessed by $K_n - e$. Equality in the one-line norm estimate is rigid: the two endpoints of the missing edge must have identical weights to all remaining vertices. For $K_n - e$ this recovers exactly the block form of the non-SSPWL class in [3].

After trace normalization, the same inequality also yields a short bound for Fiedler's absolute algebraic connectivity. The optimized bound is already contained in the extremal results of Kirkland–Pati and, in unit-budget form for $K_n - e$, in Gomyou's computation. Our point here is different: (1) holds for every individual weighted Laplacian, and equality determines its matrix block form.

For the one-missing-edge graph, the obstruction describes the entire closed cone in every order.

Theorem 1.1 (Exact closed region for $K_n - e$)**.** *For every $n \geq 3$,*

$$\overline{\mathcal{E}_{>0}(K_n - e)} = \mathcal{E}_{\geq 0}(K_n - e) = \left\{(\lambda_2, \ldots, \lambda_n) : 0 \leq \lambda_2 \leq \cdots \leq \lambda_n,\ \lambda_n \geq \frac{n}{n-2}\lambda_2\right\}. \tag{2}$$

The new sufficiency mechanism is global. Starting from a strict complete-graph realization, we pass to

$$B = I - \frac{1}{\lambda_n}L$$

and move along its fixed-$\mathbf{1}$ isospectral orbit. The extreme inequality is exactly the threshold at which that orbit can reach a zero off-diagonal entry. First contact with the entrywise nonnegative cone produces a relaxed $K_n - e$ realization. Several entries can vanish simultaneously at an arbitrary first-contact point. The next theorem shows that one can always choose a first contact with exactly one vanishing entry once $n \geq 5$.

Theorem 1.2 (Exact strict support for $K_n - e$)**.** *For every $n \geq 5$,*

$$\mathcal{E}_{>0}(K_n - e) = \left\{(\lambda_2, \ldots, \lambda_n) : 0 < \lambda_2 \leq \cdots \leq \lambda_n,\ \lambda_n \geq \frac{n}{n-2}\lambda_2\right\}.$$

The proof is global and includes repeated spectra. Inside the fixed-spectrum orthogonal orbit, the boundary of the chamber with all off-diagonal entries positive has codimension one. For each pair of entries, the locus on which both vanish has codimension at least two. The latter statement follows from a commutator criterion and an explicit analysis of the two pencils associated with adjacent and disjoint edge pairs. Hence the chamber boundary contains a point with exactly one zero. The equality face is supplied separately by the rigid block construction.

Together with the low-order analysis, the same mechanism gives a complete all-order prescribed-support PSD–SDIEP slice; see Corollary 6.4.

At order four, strict support has one exceptional equality ray.

Theorem 1.3 (Exact region for $K_4 - e$)**.** *Let $0 < x \le y \le z$. There exists $L \in \mathcal{S}_L(K_4 - e)$ with spectrum $\{0, x, y, z\}$ if and only if*

$$z \ge 2x$$

and $(x, y, z) \ne (t, t, 2t)$ for every $t > 0$. Equivalently,

$$\overline{\mathcal{E}_{>0}(K_4 - e)} = \{(x, y, z) : 0 \le x \le y \le z,\ z \ge 2x\}.$$

Combining this order-four theorem with the $n \ge 5$ strict cone and the path case shows that the missing-edge half-space is also the exact union of the strict spectral regions over all connected noncomplete graphs of a fixed order; see Corollary 6.3.

The repeated-eigenvalue result of [6] is recovered on both slices $x = y$ and $y = z$. The result also distinguishes two weak loci found in [3]: $\lambda_4 = 2\lambda_2$ is the outer boundary, whereas much of $\lambda_3 = 2\lambda_2$ lies in the interior.

For strict support in every order, we prove a more flexible insertion theorem. Put $\rho_n = n/(n-2)$.

Theorem 1.4 (Distinguished spectral-pair insertion)**.** *Let $n \ge 3$, let Λ be a multiset of $n - 1$ positive numbers, and distinguish two occurrences $0 < \alpha \le \beta$ in Λ. If*

$$\beta \ge \rho_n \alpha, \qquad t > \frac{\alpha + \beta}{n - 1} \quad \text{for every remaining occurrence } t,$$

then $\{0\} \cup \Lambda$ is the spectrum of a strictly weighted $K_n - e$ Laplacian.

Taking $(\alpha, \beta) = (\lambda_2, \lambda_3)$ recovers the simpler sufficient cone $\lambda_3 \ge \rho_n \lambda_2$. The insertion theorem, together with a separate rank-one construction for large spikes, also gives a complete strict classification on an infinite family of repeated-eigenvalue rays.

Theorem 1.5 (One-spike spectra)**.** *Let $0 < a \le b$. The spectrum $\{0, a^{(n-2)}, b\}$ has a strict $K_n - e$ realization exactly in the following cases:*

n	*condition*
3	$b \ge 3a$,
4	$b > 2a$,
$n \ge 5$	$b \ge \dfrac{n}{n-2} a$.

For $n = 4$ and $b = 2a$, the spectrum belongs to the closure but has no strict $K_4 - e$ realization.

The first higher-defect frontier is already nonlinear. If the two missing edges form a matching, all real linear witnesses supported on those two nonedges give, in the ordered chamber, only

$$\lambda_n + \frac{n-4}{n} \lambda_{n-1} \ge \lambda_2 + \lambda_3. \tag{3}$$

Theorem 8.2 proves this collapse. The facet is not asserted to characterize the full cone, but it is sufficient on one complete two-level slice: Theorem 8.3 gives

$$\{0, a^{(n-3)}, b^{(2)}\} \in \mathcal{E}_{\ge 0}(K_n - M_2) \iff b/a \ge n/(n-2),$$

and

$$\{0, a^{(n-3)}, b^{(2)}\} \in \mathcal{E}_{>0}(K_n - M_2) \iff \begin{cases} b/a - 1 \geq 2/(n-2), & n \geq 7, \\ b/a - 1 > 1/2, & n = 6. \end{cases}$$

At the excluded order-six endpoint, Ky Fan equality forces a third missing edge. Thus the complement-orbitope linear method is exhausted, while an exact nonlinear support result survives on the first rank-two slice.

The exact closed cone and the exponential representation of uniform simplex measure give

$$P(K_n - e) = 1 - \binom{(n-2)(n+1)/2}{n-2}^{-1}. \tag{4}$$

In particular, $P(K_n - e) \to 1$ with an explicit factorial-scale gap, hence faster than e^{-cn} for every fixed $c > 0$.

Erchenko, Jakobson, and Tsypin introduced $P(G)$ and asked about the volume gain under edge addition and about repeated suspension [5, Questions 3–5]. Since $P(K_n) = 1$, (4) gives

$$P(K_n) - P(K_n - e) = \binom{(n-2)(n+1)/2}{n-2}^{-1}.$$

Thus no order-independent positive lower bound can hold for the last-edge gain, under the meaningful connected interpretation of their question. This does not settle the fixed-order connected interpretation for general n.

Our first asymptotic theorem removes the special form $K_n - e$ altogether. Recall that the join $G \vee H$ is obtained from the disjoint union of G and H by adding every edge between the two vertex sets.

Theorem 1.6 (Iterated-suspension flexibility)**.** *For every nonempty finite graph G there are constants $C_G, c_G > 0$ such that*

$$P(G \vee K_k) \geq 1 - C_G e^{-c_G k} \qquad (k \geq 1). \tag{5}$$

Consequently, if $G_{k+1} = \mathrm{susp}(G_k)$, then

$$\lim_{k \to \infty} P(G_k) = 1.$$

This gives an affirmative answer to [5, Question 5], with the stronger exponential rate. The mechanism is probabilistic rather than a further deformation of the $K_n - e$ construction. Local inverse-spectral openness provides a fixed finite spectral template for G; the weighted join formula shifts that template into the target list; and exponential order statistics find the required template among a growing number of independent spacings. The same argument gives a uniform theorem for every graph obtained from K_n by deleting at most a fixed number of edges.

For the smallest defects we obtain the sharp scale, which is much faster than the preceding exponential estimate.

Theorem 1.7 (Optimal one- and two-edge defect law)**.** *Let G_n be obtained from K_n by deleting exactly one or exactly two edges. Then, as $n \to \infty$ and uniformly over the two isomorphism types when two edges are deleted,*

$$1 - P(G_n) = \exp\{-n \log n + O(n)\}. \tag{6}$$

Here M_2 denotes the two-edge matching and $E(P_3)$ a pair of adjacent edges; these give $K_n - e$, $K_n - M_2$, and $K_n - E(P_3)$.

For the two-edge cases, the new ingredient is a moving anchor in the join decomposition. A single fixed spectral template yields only an exponentially small failure bound. Moving the distinguished join eigenvalue through the first $\lfloor n/\log n \rfloor$ order statistics forces every failed chart to compress all but $O(n/\log n)$ exponentials into a window of relative width $O(1/n)$. The probability of that clustering is $\exp\{-n\log n + O(n)\}$. The missing-edge inequality supplies the matching lower bound. The two possible two-edge defects are respectively

$$K_n - M_2 = C_4 \vee K_{n-4}, \qquad K_n - E(P_3) = \mathrm{Paw} \vee K_{n-4}.$$

At order four, direct integration of the exact cones yields the complete volume census and the sharp value $c_4^{\mathrm{conn}} = 1/10$ for connected one-edge additions. The main contributions of this paper are the pointwise missing-edge inequality and equality-support rigidity, the exact strict $K_n - e$ cone and universal noncomplete union theorem, the single-contact orbit theorem, the all-order $K_n - e$-support PSD–SDIEP slice, the spectral-pair and one-spike constructions, the collapse of the M_2 complement-orbitope family, the exact top-rank-two M_2 slice, the exact one-edge volume, the fixed-core volume theorem, and the moving-anchor two-edge defect law. The join identity, local spectral openness, and the optimized absolute-algebraic-connectivity bound are used with explicit attribution and are not claimed as new.

Section 2 fixes strict and relaxed support conventions. Sections 3–6 prove the obstruction, the exact closed cone, exact strict support, and the exact four-vertex result. Section 7 gives the strict spectral-pair constructions, and Section 8 treats the two-edge matching frontier. Sections 9–11 establish the volume formula, suspension theorem, and sharp defect law. Section 12 records the four-vertex census and edge-addition consequences, and Section 13 isolates the remaining edge-addition, suspension, and finite-defect problems.

2. Weighted Laplacians and spectral volume

Let $G = (V, E)$ be a simple graph with $V = \{1, \dots, n\}$. A strictly weighted Laplacian on G is the symmetric matrix $L = (L_{ij})$ determined by weights $w_{ij} > 0$ on $ij \in E$:

$$L_{ij} = \begin{cases} -w_{ij}, & i \neq j,\ ij \in E, \\ 0, & i \neq j,\ ij \notin E, \end{cases} \qquad L_{ii} = \sum_{j:ij\in E} w_{ij}.$$

We write $\mathcal{S}_L(G)$ for this class. Every $L \in \mathcal{S}_L(G)$ is positive semidefinite and satisfies $L\mathbf{1} = 0$. If G is connected, its ordered spectrum is

$$0 = \lambda_1 < \lambda_2 \leq \cdots \leq \lambda_n.$$

The closure $\overline{\mathcal{S}_L(G)}$ permits zero weights on allowed edges. This distinction matters for exact realization at lower-dimensional faces but not for Euclidean volume.

We use

$$H_n = I_n - \frac{1}{n} J_n$$

for the orthogonal projection onto $\mathbf{1}^{\perp}$. When the dimension is clear we write simply H.

For completeness, we fix the volume convention used later. Put $m = n - 1$ and let

$$\Delta_m = \left\{ (t_1, \dots, t_m) \in \mathbb{R}^m_{\geq 0} : \sum_{i=1}^{m} t_i = 1 \right\}.$$

Let $\widehat{\mathcal{R}}(G) \subseteq \Delta_m$ be the union of all coordinate permutations of the normalized $n-1$ spectral coordinates obtained after deleting the distinguished zero eigenvalue; in the relaxed class these coordinates may contain additional zeros. Following [5, Section 8], define

$$P(G) = \frac{\mathrm{vol}_{m-1}(\widehat{\mathcal{R}}(G))}{\mathrm{vol}_{m-1}(\Delta_m)}. \tag{7}$$

To make the two support conventions precise, let $\mathcal{S}_L^{>0}(G) = \mathcal{S}_L(G)$ and let $\mathcal{S}_L^{\geq 0}(G)$ permit zero weights on allowed edges. Define the ordered spectral sets by

$$\mathcal{E}_\sigma(G) = \{(\lambda_2, \dots, \lambda_n) : \exists L \in \mathcal{S}_L^\sigma(G),\ \mathrm{Spec}(L) = \{0, \lambda_2, \dots, \lambda_n\},\ 0 \leq \lambda_2 \leq \cdots \leq \lambda_n\},$$

where σ is either > 0 or ≥ 0.

Proposition 2.1 (Strict and relaxed support)**.** *For every graph G,*

$$\overline{\mathcal{E}_{>0}(G)} = \mathcal{E}_{\geq 0}(G).$$

After trace normalization, the strict and relaxed regions have the same Euclidean volume.

Proof. Given a nonnegative weighting, add $\varepsilon > 0$ to every zero allowed edge weight. The resulting Laplacians lie in $\mathcal{S}_L^{>0}(G)$ and converge to the original matrix, so their ordered spectra converge as well. Conversely, if a sequence of spectra converges, then its traces are bounded. Since the trace of a weighted Laplacian is twice the sum of its edge weights, all weights are bounded; a convergent subsequence therefore has a nonnegative limiting weighting. After trace normalization, the spectral images are semialgebraic. For every semialgebraic set S one has

$$\dim(\overline{S} \setminus S) < \dim S.$$

Thus closure adds only lower-dimensional strata and does not change ambient simplex volume; if S is already lower-dimensional, both ambient volumes are zero. □

We use the relaxed region in (7), matching the volume convention of [5], and the strict region for exact-support realization theorems.

3. A sharp obstruction from one missing edge

The next estimate applies to every noncomplete graph, not only to $K_n - e$.

Theorem 3.1 (Missing-edge inequality)**.** *Let G be a noncomplete connected graph on $n \geq 3$ vertices. If $L \in \mathcal{S}_L(G)$ has ordered spectrum*

$$0 = \lambda_1 < \lambda_2 \leq \cdots \leq \lambda_n,$$

then

$$\lambda_n \geq \frac{n}{n-2}\lambda_2. \tag{8}$$

The universal constant is sharp in every order, as witnessed by $K_n - e$.

Proof. Choose a missing edge and label its endpoints $1, 2$. Set

$$x = \lambda_2, \qquad z = \lambda_n, \qquad m = \frac{x+z}{2}, \qquad \delta = \frac{z-x}{2},$$

and define $T = L - mH_n$. On $\mathbf{1}^\perp$, every eigenvalue of T lies in $[-\delta, \delta]$, hence

$$\|T|_{\mathbf{1}^\perp}\| \leq \delta.$$

Put $p = H_n e_1$ and $q = H_n e_2$. Then

$$\|p\| = \|q\| = \sqrt{\frac{n-1}{n}}, \qquad (H_n)_{12} = -\frac{1}{n}.$$

Because $L_{12} = 0$ and $T\mathbf{1} = 0$,

$$p^\mathsf{T} T q = e_1^\mathsf{T} T e_2 = \frac{m}{n}.$$

Therefore

$$\frac{m}{n} \leq \delta \|p\| \|q\| = \delta \frac{n-1}{n}.$$

Substituting m and δ gives $nx \leq (n-2)z$, as required. Sharpness follows from Proposition 3.2 below. □

Equality in Theorem 3.1 remembers much more than the two extreme eigenvalues.

Proposition 3.2 (Equality rigidity)**.** *Let G be a noncomplete connected graph on $n \geq 3$ vertices, let $12 \notin E(G)$, and let $L \in \mathcal{S}_L(G)$ have ordered spectrum*

$$0 = \lambda_1 < \lambda_2 \leq \cdots \leq \lambda_n.$$

Suppose equality holds in (8). Put $s = n - 2$ and

$$\alpha = \frac{\lambda_2}{s} = \frac{\lambda_n}{n}.$$

After ordering the vertices as $1, 3, \ldots, n, 2$, the matrix L has the form

$$L = \begin{pmatrix} s\alpha & -\alpha \mathbf{1}_s^{\mathsf{T}} & 0 \\ -\alpha \mathbf{1}_s & B + 2\alpha I_s & -\alpha \mathbf{1}_s \\ 0 & -\alpha \mathbf{1}_s^{\mathsf{T}} & s\alpha \end{pmatrix}, \tag{9}$$

where B is a weighted Laplacian on the induced graph on the other s vertices. If the ordered eigenvalues of B are $0 = \mu_1 \leq \mu_2 \leq \cdots \leq \mu_s$, then

$$\operatorname{Spec}(L) = \{0, s\alpha, n\alpha, \mu_2 + 2\alpha, \ldots, \mu_s + 2\alpha\}. \tag{10}$$

When $s \geq 2$, the equality hypothesis also forces

$$(s-2)\alpha \leq \mu_2 \leq \cdots \leq \mu_s \leq s\alpha. \tag{11}$$

Conversely, every matrix of the form (9) with $\alpha > 0$ and a nonnegatively weighted Laplacian B satisfying (11) when $s \geq 2$ has extreme positive eigenvalues $s\alpha, n\alpha$ and attains equality. In particular, equality is attained by a strictly weighted $K_n - e$ for every $n \geq 3$.

Proof. Use the notation in the proof of Theorem 3.1. Equality in

$$p^{\mathsf{T}} T q \leq \|p\| \, \|Tq\| \leq \delta \|p\| \, \|q\|,$$

together with $\|p\| = \|q\|$, forces $Tq = \delta p$. Since T is self-adjoint and $\|T\| \leq \delta$, norm saturation gives $T^2 q = \delta^2 q$, and hence $Tp = \delta q$. It follows that

$$e_1 - e_2 = p - q$$

is an x-eigenvector, while

$$e_1 + e_2 - \frac{2}{n}\mathbf{1} = p + q$$

is a z-eigenvector. Looking at a coordinate $k \notin \{1, 2\}$ in the first identity gives $L_{k1} = L_{k2}$. Looking at the same coordinate in the second identity gives

$$L_{k1} + L_{k2} = -\frac{2z}{n}.$$

Thus every edge from an endpoint of the missing edge to a remaining vertex has weight $z/n = \alpha$. Subtracting these two incident weights from each central diagonal leaves a weighted Laplacian B, proving (9).

The spectrum in (10) follows from the orthogonal invariant decomposition

$$\mathbb{R}^n = \langle \mathbf{1} \rangle \oplus \langle e_1 - e_2 \rangle \oplus \langle (s, -2\mathbf{1}_s, s) \rangle \oplus \left(\{0\} \oplus \mathbf{1}_s^{\perp} \oplus \{0\} \right).$$

The shifted central eigenvalues must lie between the two extremes $s\alpha$ and $n\alpha$, which is exactly (11). The same decomposition proves the stated converse.

For sharpness, the case $n = 3$ is the equally weighted path. For $n \geq 4$, choose $0 < \varepsilon < 2\alpha$ and let B be the equally weighted complete graph Laplacian on s vertices with every nonzero eigenvalue equal to

$$\mu = (s-2)\alpha + \varepsilon > 0.$$

Then every shifted central eigenvalue is $s\alpha+\varepsilon$, strictly between $s\alpha$ and $n\alpha$. Hence the ordered extreme eigenvalues are $s\alpha$ and $n\alpha$, and equality holds. □

Corollary 3.3 (Support rigidity on the universal facet)**.** *Let G be a connected noncomplete graph on $n\geq 3$ vertices and let $L\in\mathcal{S}_L(G)$ attain equality in (8). Then $\overline{G}$, the graph complement of G, is a nonempty matching. More precisely, there is a single $\alpha>0$ such that, for every nonedge uv and every $k\notin\{u,v\}$,*

$$w_{uk}=w_{vk}=\alpha.$$

Conversely, if M_r denotes a matching of r independent edges on the n-vertex set, where $1\leq r\leq\lfloor n/2\rfloor$, then assigning the uniform weight α to every edge of K_n-M_r gives

$$\operatorname{Spec}L=\{0,((n-2)\alpha)^{(r)},(n\alpha)^{(n-1-r)}\}. \tag{12}$$

Thus every matching defect attains the sharp universal bound along the displayed ray.

Proof. Apply Proposition 3.2 to each nonedge uv. It makes every edge from u or v to a third vertex present with weight α; in particular, two nonedges cannot share an endpoint. The value $\alpha=\lambda_2/(n-2)=\lambda_n/n$ is independent of the chosen nonedge.

Conversely, if the missing pairs are u_jv_j, then

$$L=\alpha\left(nH_n-\sum_{j=1}^{r}(e_{u_j}-e_{v_j})(e_{u_j}-e_{v_j})^{\mathsf{T}}\right).$$

The r displayed difference vectors are mutually orthogonal and lie in $\mathbf{1}^{\perp}$. They have eigenvalue $(n-2)\alpha$; their orthogonal complement in $\mathbf{1}^{\perp}$ has eigenvalue $n\alpha$. This proves (12). □

For $n\geq 4$, the block form (9) is precisely the class that fails the strong spectral property for weighted Laplacians, as classified by Catral, Fallat, Gupta, and Lin [3, Theorem 4.2]. Combining their classification with the complete-graph realization theorem of [6, Theorem 2.5] gives the following useful reformulation. A positive multiset Λ admits such a non-SSPWL realization in $\mathcal{S}_L(K_n-e)$ if and only if it contains distinguished elements u,v such that

$$v=\frac{n}{n-2}u,\qquad t>\frac{2}{n-2}u\quad(t\in\Lambda\setminus\{u,v\}). \tag{13}$$

This is a corollary of the cited results rather than a new classification.

We will also use the following consequence of the block form.

Corollary 3.4 (The equality face of K_n-e)**.** *Let $0<x\leq t_1\leq\cdots\leq t_{n-3}\leq z$ and $z=\frac{n}{n-2}x$. For $n\geq 5$, the spectrum*

$$\{0,x,t_1,\dots,t_{n-3},z\}$$

is realized by a strictly weighted K_n-e. For $n=4$ it is realized if and only if $t_1>x$; the ray $(x,x,2x)$ belongs only to the closure. For $n=3$, the equality ray is realized by P_3.

Proof. Set $s=n-2$ and $\alpha=x/s$. In (10) prescribe the nonzero spectrum of $B\in\mathcal{S}_L(K_s)$ to be

$$t_i-2\alpha\qquad(1\leq i\leq s-1).$$

For $n\geq 5$ one has $2\alpha<x\leq t_i$, so all these numbers are positive and the complete-graph theorem [6, Theorem 2.5] supplies B. When $n=4$, the single shifted eigenvalue is t_1-x, which is positive exactly when $t_1>x$. The case $n=3$ is immediate. □

3.1. **Absolute algebraic connectivity.** For an edge weighting $w \in \mathbb{R}_{\geq 0}^{E(G)}$, write

$$L_G(w) = \sum_{ij \in E(G)} w_{ij}(e_i - e_j)(e_i - e_j)^{\mathsf{T}}.$$

Following Fiedler's edge-budget formulation [7]; see also [8] for trees, the absolute algebraic connectivity of a graph G with $M = |E(G)|$ is

$$\widehat{a}(G) = \max \left\{ \lambda_2(L_G(w)) : w_{ij} \geq 0, \quad \sum_{ij \in E(G)} w_{ij} = M \right\}. \tag{14}$$

Thus the average edge weight is one and $\operatorname{tr} L_G(w) = 2M$. The feasible simplex is compact. Its relative interior is dense, so continuity shows that replacing $w_{ij} \geq 0$ by $w_{ij} > 0$ leaves the supremum unchanged, although that strict supremum need not be attained. General bounds involving vertex connectivity were obtained by Kirkland and Pati [11].

Corollary 3.5 (Trace consequence of the missing-edge inequality)**.** *Let G be a connected noncomplete graph on $n \geq 3$ vertices, and put $M = |E(G)|$. Then*

$$\widehat{a}(G) \leq \frac{2M(n-2)}{n^2 - 3n + 4}. \tag{15}$$

Proof. Let $L = L_G(w)$ be feasible in (14), with ordered spectrum $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$. If $\lambda_2 = 0$, there is nothing to prove. Otherwise the positive support of w is connected. The proof of Theorem 3.1 applies unchanged to nonnegative weights, or equivalently by continuity from strict weights, and hence

$$\lambda_n \geq \frac{n}{n-2}\lambda_2.$$

The trace normalization now gives

$$\begin{aligned} 2M &= \sum_{i=2}^{n} \lambda_i \\ &\geq (n-2)\lambda_2 + \lambda_n \\ &\geq \left(n - 2 + \frac{n}{n-2}\right)\lambda_2 = \frac{n^2 - 3n + 4}{n-2}\lambda_2. \end{aligned}$$

Taking the maximum proves (15). □

The optimized bound in Corollary 3.5 is not new: it is contained in the absolute-connectivity extremal results of Kirkland and Pati [11]. For $n \geq 5$, it also follows in unit-budget normalization from Gomyou's computation for $K_n - e$ [10, Proposition 3.7]. Indeed, let $\rho(G)$ denote the optimum under the unit budget $\sum_e w_e = 1$, so that $\widehat{a}(G) = M\rho(G)$. Every noncomplete n-vertex graph is a subgraph of some $K_n - e$, and ρ is nondecreasing when allowed edges are added because new edges may receive weight zero. Gomyou's value is $\rho(K_n - e) = 2(n-2)/(n^2 - 3n + 4)$, which gives (15) after multiplication by M. The cases $n = 3, 4$ follow directly. Thus the role of the corollary is to contrast that optimization statement with the pointwise inequality (8) and its equality rigidity.

For completeness, put $D_n = n^2 - 3n + 4$. In Fiedler's normalization,

$$\widehat{a}(K_n - e) = \frac{(n-2)^2(n+1)}{D_n}. \tag{16}$$

If the missing edge is 12, an optimal relaxed weighting is

$$w_{1j} = w_{2j} = \frac{(n-2)(n+1)}{D_n}, \qquad 3 \le j \le n, \tag{17}$$

$$w_{ij} = \frac{(n-4)(n+1)}{D_n}, \qquad 3 \le i < j \le n. \tag{18}$$

For $n = 3$ the second index family is empty, and the construction is the unit-weighted path P_3. Its spectrum is

$$\left\{0,\ \left(\frac{(n-2)^2(n+1)}{D_n}\right)^{(n-2)},\ \frac{n(n-2)(n+1)}{D_n}\right\}. \tag{19}$$

Indeed, if $s = n-2$ and a, b denote the central and cross weights above, then

$$\binom{s}{2}a + 2sb = \binom{n}{2} - 1, \qquad sa + 2b = sb.$$

The endpoint/core decomposition from Proposition 3.2 therefore gives (19); equality in (15) proves optimality. For $n \ge 5$ all allowed weights are positive. For $K_4 - e$ the central weight is zero, so $\widehat{a}(K_4 - e) = 5/2$ is attained in the relaxed problem but is only a supremum over strict $K_4 - e$ weightings, in agreement with [3, Section 6].

4. The exact closed cone for $K_n - e$

We now prove Theorem 1.1. The decisive step is a first-contact statement for a fixed-$\mathbf{1}$ orthogonal orbit.

Lemma 4.1 (First contact with the nonnegative cone)**.** *Let*

$$1 = \nu_1 > \nu_2 \ge \cdots \ge \nu_n = 0, \qquad n \ge 3,$$

put $D_\nu = \operatorname{diag}(\nu_2, \dots, \nu_n)$*, and consider the orbit*

$$\mathcal{O}_\nu = \left\{\frac{1}{n}J_n + QD_\nu Q^{\mathsf{T}} : Q^{\mathsf{T}}Q = I_{n-1},\ Q^{\mathsf{T}}\mathbf{1} = 0\right\}.$$

Assume that $\mathcal{O}_\nu$ *contains a matrix whose off-diagonal entries are all strictly positive. Then* $\mathcal{O}_\nu$ *contains an entrywise nonnegative matrix having a zero off-diagonal entry if and only if*

$$\nu_2 \ge \frac{2}{n}. \tag{20}$$

Moreover, for every fixed pair $i \ne j$*,*

$$\min_{B \in \mathcal{O}_\nu} B_{ij} = \frac{1}{n} - \frac{\nu_2}{2}. \tag{21}$$

Proof. Fix distinct indices i, j and put $p_i = H_n e_i$ and $p_j = H_n e_j$. The vectors

$$u = \frac{p_i + p_j}{\sqrt{2(n-2)/n}}, \qquad v = \frac{p_i - p_j}{\sqrt{2}}$$

are orthonormal, and

$$p_i = \sqrt{\frac{n-2}{2n}}u + \frac{1}{\sqrt{2}}v, \qquad p_j = \sqrt{\frac{n-2}{2n}}u - \frac{1}{\sqrt{2}}v.$$

Write $B = J_n/n + C$ for a member of $\mathcal{O}_\nu$. On $\mathbf{1}^\perp$ the eigenvalues of C are $\nu_2, \dots, \nu_n$, and hence

$$0 \preceq C \preceq \nu_2 H_n.$$

The mixed terms cancel, giving

$$B_{ij} = \frac{1}{n} + \frac{n-2}{2n} u^{\mathsf{T}} C u - \frac{1}{2} v^{\mathsf{T}} C v \geq \frac{1}{n} - \frac{\nu_2}{2}. \tag{22}$$

Thus a nonnegative orbit point with a zero off-diagonal entry can exist only if (20) holds.

The bound is attained for the fixed pair i, j. Orient the $\nu_n = 0$ eigendirection of C along u, the ν_2 eigendirection along v, and distribute the remaining eigendirections on $\{u, v\}^{\perp} \cap \mathbf{1}^{\perp}$. The resulting $B_1 \in \mathcal{O}_\nu$ satisfies

$$(B_1)_{ij} = \frac{1}{n} - \frac{\nu_2}{2}.$$

Together with (22), this proves (21); under (20) this value is nonpositive.

Let $B_0 \in \mathcal{O}_\nu$ have all off-diagonal entries strictly positive. The orbit is path connected, including in the presence of repeated eigenvalues. Indeed, after fixing an orthonormal basis of $\mathbf{1}^{\perp}$, every orbit point has a representative $RD_\nu R^{\mathsf{T}}$ with $R \in O(n-1)$. Multiplying R on the right by a determinant--1 diagonal sign matrix commuting with D_ν changes the determinant without changing the represented matrix. Thus every orbit point has an $SO(n-1)$ representative, and $SO(n-1)$ is path connected for $n \geq 3$.

Choose a continuous path $B(t)$ from B_0 to B_1 inside the orbit and set

$$f(t) = \min_{r<s} B_{rs}(t), \qquad t_* = \min\{t \in [0,1] : f(t) \leq 0\}.$$

Continuity gives $f(t_*) = 0$. Hence every off-diagonal entry of $B(t_*)$ is nonnegative and at least one is zero. Every orbit point is positive semidefinite, so its diagonal entries are nonnegative as well. Therefore $B(t_*)$ is the required entrywise nonnegative first-contact matrix. □

Proof of Theorem 1.1. Consider first a relaxed $K_n - e$ Laplacian. If $\lambda_2 = 0$, the required inequality is automatic. If $\lambda_2 > 0$, let H be the graph of positive-weight edges. Then H is connected, $H \subseteq K_n - e$ is noncomplete, and $L \in \mathcal{S}_L(H)$. Theorem 3.1 therefore applies and proves necessity.

For sufficiency, first suppose $0 < \lambda_2 \leq \cdots \leq \lambda_n$ and define

$$\nu_i = 1 - \frac{\lambda_i}{\lambda_n} \qquad (2 \leq i \leq n).$$

Then

$$1 > \nu_2 \geq \cdots \geq \nu_n = 0, \qquad \lambda_n \geq \frac{n}{n-2}\lambda_2 \iff \nu_2 \geq \frac{2}{n}.$$

By complete-graph flexibility [6, Theorem 2.5], there is a strictly weighted complete-graph Laplacian L_0 with spectrum $\{0, \lambda_2, \ldots, \lambda_n\}$. The matrix

$$B_0 = I - \frac{1}{\lambda_n} L_0$$

belongs to $\mathcal{O}_\nu$ and has every off-diagonal entry strictly positive. Lemma 4.1 supplies an entrywise nonnegative $B_* \in \mathcal{O}_\nu$ with a zero off-diagonal entry. Relabel vertices so that $(B_*)_{12} = 0$, and put

$$L_* = \lambda_n (I - B_*).$$

Because $B_* \mathbf{1} = \mathbf{1}$, the matrix L_* has row sums zero. Its off-diagonal entries are nonpositive, $(L_*)_{12} = 0$, and its spectrum is $\{0, \lambda_2, \ldots, \lambda_n\}$. Thus L_* is a nonnegatively weighted Laplacian supported on $K_n - e$; additional zero allowed edges are permitted in the relaxed class.

It remains to include lists with zero eigenvalues. We are now in the case $\lambda_2 = 0$. If $\lambda_n = 0$, the target is the zero tuple and is realized by the zero weighting. Otherwise, for sufficiently small $\varepsilon > 0$ put

$$\mu_i^{(\varepsilon)} = \lambda_i + \varepsilon \qquad (2 \leq i \leq n).$$

The list remains ordered and, provided

$$\varepsilon \le \frac{\lambda_n}{\frac{n}{n-2}-1},$$

satisfies

$$\mu_n^{(\varepsilon)} \ge \frac{n}{n-2}\mu_2^{(\varepsilon)}.$$

Choose a sequence $\varepsilon_k \downarrow 0$. The positive case gives a relaxed realization for each ε_k. Their traces are uniformly bounded and every edge weight is at most half the trace, so the weight vectors have a convergent subsequence. Its limit realizes the original list. Proposition 2.1 then gives

$$\mathcal{E}_{\ge 0}(K_n - e) = \overline{\mathcal{E}_{>0}(K_n - e)},$$

completing the proof. □

5. Single contact and exact strict support

We now remove the support ambiguity in the first-contact argument. The key point is that a hypersurface can separate an orthogonal orbit, whereas the union of its double-contact strata cannot.

Lemma 5.1 (Fixed-locus codimension)**.** *Let V have dimension N, let $\mathcal{O}$ be a nonscalar real-symmetric orthogonal orbit on V, and let T be symmetric with eigenspace dimensions $k_1, \dots, k_s$. If $k = \max_j k_j < N$, then every component of*

$$\{C \in \mathcal{O} : [C, T] = 0\}$$

has codimension at least $N - k$ in $\mathcal{O}$.

Proof. Suppose that the distinct eigenvalues on $\mathcal{O}$ have multiplicities $m_1, \dots, m_r$. A component of the fixed locus is specified by a table (r_{ij}) whose row sums are m_i and whose column sums are k_j. Its dimension and the orbit dimension are respectively

$$\sum_j \sum_{i<i'} r_{ij} r_{i'j}, \qquad d = \sum_{i<i'} m_i m_{i'}.$$

Thus the codimension counts the unordered pairs of coordinate directions that lie in different C-eigenblocks and different T-eigenblocks. Choose a nontrivial C-eigenblock and let a_j be the dimension of its intersection with the jth T-eigenspace; put $p = \sum_j a_j$, so $0 < p < N$. The pairs crossing from this block to its complement and lying in different T-eigenspaces form a cut in the complete multipartite graph with part sizes $k_1, \dots, k_s$. Its size is

$$p(N-p) - \sum_j a_j(k_j - a_j).$$

The edge connectivity of this complete multipartite graph is its minimum degree $N - k$. Directly, minimizing the displayed integer expression over $0 \le a_j \le k_j$ and $1 \le p \le N-1$ gives $N - k$, attained by isolating one vertex in a part of size k. All these cut pairs occur in the full codimension count, proving the asserted bound. □

Lemma 5.2 (Double-contact strata)**.** *Let $n \ge 5$, let*

$$1 > \nu_2 \ge \cdots \ge \nu_n = 0, \qquad \nu_2 > \frac{2}{n},$$

and let $\mathcal{O}_\nu$ be the orbit in Lemma 4.1. For two distinct off-diagonal positions e and f, the subset of $\mathcal{O}_\nu$ on which both entries vanish has codimension at least two.

Proof. Put $N = n-1$, $V = \mathbf{1}^{\perp}$, and write $B = J_n/n + C$. If the distinct eigenvalues of $C|_V$ have multiplicities $m_1, \dots, m_r$, then

$$d := \dim \mathcal{O}_\nu = \frac{N^2 - \sum_i m_i^2}{2} = \sum_{i<i'} m_i m_{i'}. \tag{23}$$

The operator is nonscalar, so $d \geq N - 1 \geq 3$.

For $e = ij$, set

$$X_e = \frac{1}{2}(e_i e_j^{\mathsf{T}} + e_j e_i^{\mathsf{T}}), \qquad S_e = H_n X_e H_n|_V, \qquad f_e(B) = \frac{1}{n} + \langle C, S_e \rangle = B_{ij},$$

where $\langle A, D \rangle = \operatorname{tr}(AD)$. Since $T_C \mathcal{O}_\nu = \{[K, C] : K^{\mathsf{T}} = -K\}$, one has

$$a\, df_e + b\, df_f = 0 \quad \Longleftrightarrow \quad [C, aS_e + bS_f] = 0. \tag{24}$$

At points where (df_e, df_f) has rank two, the common zero set has local codimension two. It remains to control the dependent locus. Up to a vertex permutation there are two cases.

Suppose first that $e = 12$ and $f = 13$. For $T = aS_{12} + bS_{13} \neq 0$, direct restriction to V shows that T has rank two, with nonzero eigenvalues $t_+ > 0 > t_-$ satisfying

$$\operatorname{tr} T = -\frac{a+b}{n}, \qquad t_+ t_- = -\frac{Q(a,b)}{4n}, \qquad Q(a,b) = (n-2)(a^2+b^2) - 2ab > 0. \tag{25}$$

If C commutes with T, its values on the two nonzero eigendirections are some c_+, c_- from the fixed finite set $\{\nu_2, \dots, \nu_n\}$. On the double zero set,

$$\operatorname{tr}(CT) = -\frac{a+b}{n} = \operatorname{tr} T,$$

and therefore

$$\frac{t_+}{-t_-} = \frac{1 - c_-}{1 - c_+} =: R > 0. \tag{26}$$

For each fixed R, equations (25) and (26) give

$$4R(a+b)^2 - n(R-1)^2 Q(a,b) = 0. \tag{27}$$

This homogeneous quadratic is not identically zero: its ab coefficient would otherwise require $8R + 2n(R-1)^2 = 0$. Hence only finitely many projective parameters $[a : b]$ occur. Each associated T has eigenspace dimensions $(N-2, 1, 1)$, so Lemma 5.1 gives codimension at least two.

Suppose next that $e = 12$ and $f = 34$. The characteristic polynomial of $T = aS_{12} + bS_{34}$ on V is

$$x^{N-4}\left(x + \frac{a}{2}\right)\left(x + \frac{b}{2}\right)\left[x^2 - \frac{(n-2)(a+b)}{2n}x + \frac{(n-4)ab}{4n}\right]. \tag{28}$$

For $ab \neq 0$ and all but finitely many $[a : b]$, the four nonzero roots are simple. The largest eigenspace then has dimension $\max(N-4, 1)$, so the fixed locus has codimension at least $\min(4, N-1) \geq 3$. Allowing the one projective parameter still leaves codimension at least two.

The exceptional projective ratios are finite: they are the axes together with the zeros of the quadratic discriminant and of the pairwise resultants of the factors in (28). At an axis, the zero eigenspace has dimension $N-2$. Away from the axes, a nonzero eigenvalue has multiplicity at most two. Indeed, if the quadratic factor is denoted by $q(x)$, then

$$q(-a/2) = \frac{a((n-1)a + (n-3)b)}{2n}, \qquad q'(-a/2) = -\frac{(3n-2)a + (n-2)b}{2n};$$

the coefficient determinant is $-2(n^2 - 4n + 2) \neq 0$, and the same argument applies to $-b/2$. Also $a = b \neq 0$ gives $q(-a/2) = a^2(n-2)/n \neq 0$. Thus every exceptional T has largest eigenspace dimension at most $\max(N-2, 2) = N-2$. Lemma 5.1 again gives codimension at least two.

To include every singular point explicitly, consider the semialgebraic commuting-pencil incidence

$$\mathcal{I}_{e,f} = \left\{ (C, [a:b]) \in \mathcal{O}_\nu \times \mathbb{RP}^1 : f_e(C) = f_f(C) = 0,\ [C, aS_e + bS_f] = 0 \right\}.$$

By (24), its projection contains the locus on which (df_e, df_f) has rank below two. In the adjacent case, (27) leaves only finitely many $[a:b]$, and every corresponding fixed locus has codimension at least two; hence $\dim \mathcal{I}_{e,f} \le d-2$. In the disjoint case, a generic projective parameter has fixed-locus codimension at least three, so adjoining the one-dimensional parameter still gives dimension at most $d-2$. The exceptional parameters are finite and each has fixed-locus codimension at least two, giving the same bound. Projection cannot increase semialgebraic dimension. The rank-two part of the double-zero set is a codimension-two submanifold, while its rank-deficient part lies in this projection. Therefore the entire double-contact set has codimension at least two. □

Proof of Theorem 1.2. Necessity is Theorem 3.1, and the equality face is Corollary 3.4. It remains to assume

$$\lambda_n > \frac{n}{n-2}\lambda_2$$

and prove strict support. Put $\nu_i = 1 - \lambda_i/\lambda_n$ and let $M = \mathcal{O}_\nu$. Define the open semialgebraic chamber

$$\mathcal{P} = \{B \in M : B_{ij} > 0 \text{ for every } i < j\}.$$

It is nonempty: a strict complete-graph realization L_0 gives $B_0 = I - L_0/\lambda_n \in \mathcal{P}$. Its complement has nonempty relative interior. Indeed, orient the ν_2 eigendirection along $v = (p_i - p_j)/\sqrt{2}$ and a zero eigendirection along $u = (p_i + p_j)/\sqrt{2(n-2)/n}$ as in Lemma 4.1; the resulting orbit point has

$$B_{ij} = \frac{1}{n} - \frac{\nu_2}{2} < 0.$$

The orbit M is a connected smooth semialgebraic manifold of dimension d. Indeed, a reflection inside any eigenspace commutes with the diagonal representative and changes determinant, so its $O(n-1)$-orbit equals its $SO(n-1)$-orbit; this also covers repeated spectra. Since both $\mathcal{P}$ and the interior of its complement are nonempty, one has

$$M \setminus \partial_M \mathcal{P} = \mathcal{P} \dot\cup \operatorname{int}_M(M \setminus \mathcal{P}).$$

Thus $\partial_M \mathcal{P}$ separates M and has dimension $d-1$: a semialgebraic subset of dimension at most $d-2$ cannot separate a connected d-manifold [2, Chapter 2]. Equivalently, after a Whitney stratification a path can be perturbed transversely to every stratum and misses all strata of codimension at least two. Lemma 5.2 says that the finite union of all double-zero strata has dimension at most $d-2$. Consequently there is a point $B_* \in \partial_M \mathcal{P}$ outside that union. Every off-diagonal entry of B_* is nonnegative, at least one is zero, and no two are zero. After relabeling, its unique zero is B_{12}. Then

$$L_* = \lambda_n(I - B_*)$$

has the prescribed spectrum, vanishes at the missing edge 12, and is strictly negative at every other off-diagonal entry. Hence $L_* \in \mathcal{S}_L(K_n - e)$, as required. □

6. The exact region for $K_4 - e$

We now prove Theorem 1.3. The construction is a rank-two perturbation of the scalar operator on $\mathbf{1}^\perp$.

Lemma 6.1 (Explicit low-rank synthesis)**.** *Let $0 < x \le y \le z$ with $z \ge 2x$, and label the missing edge 12. Unless $(x,y,z) = (t,t,2t)$, there are orthonormal vectors $q, r \in \mathbf{1}^\perp$ such that*

$$L = xH_4 + (y-x)rr^{\mathsf{T}} + (z-x)qq^{\mathsf{T}} \tag{29}$$

satisfies $L_{12} = 0$ and $L_{ij} < 0$ on every other off-diagonal position. On the exceptional ray the same construction has one additional zero entry.

The formulas and the complete sign verification are given in Appendix A.

Proof of Theorem 1.3. Necessity of $z \geq 2x$ is Theorem 3.1 with $n = 4$. If $z = 2x$, equality rigidity gives (9) with $s = 2$ and $\alpha = x/2$. The four edges joining the missing-edge endpoints to the other two vertices therefore have weight $x/2$. If the fifth allowed edge has weight w, then (10) gives

$$y = x + 2w.$$

Thus strict positivity is impossible exactly when $y = x$.

Conversely, every triple satisfying the stated conditions is realized by Lemma 6.1. Allowing limits adds the exceptional ray and the faces with $x = 0$, which proves the closure statement. □

Corollary 6.2 ($K_4 - e$ facet rigidity)**.** *Fix the missing edge 12. If a relaxed $K_4 - e$ weighting has spectrum $\{0, x, y, 2x\}$ with $0 < x \leq y \leq 2x$, then it is uniquely determined:*

$$w_{13} = w_{14} = w_{23} = w_{24} = \frac{x}{2}, \qquad w_{34} = \frac{y - x}{2}.$$

It has exact $K_4 - e$ support if and only if $y > x$. At $y = x$, its positive support is exactly the uniformly weighted cycle C_4.

Proof. Since $x > 0$, the graph H of positive-weight edges is connected; regarding L as an element of $\mathcal{S}_L(H)$, Proposition 3.2 fixes the four endpoint–core weights at $x/2$. The central block has a single edge of weight w_{34}, and (10) gives $y = x + 2w_{34}$. These identities prove uniqueness and the support claims; conversely, the displayed weights have the stated spectrum by the same block decomposition. □

Corollary 6.3 (The universal noncomplete spectral cone)**.** *For every $n \geq 3$,*

$$\bigcup_{\substack{G \text{ connected and noncomplete} \\ |V(G)|=n}} \mathcal{E}_{>0}(G) = \left\{ 0 < \lambda_2 \leq \cdots \leq \lambda_n : \lambda_n \geq \frac{n}{n-2}\lambda_2 \right\}. \tag{30}$$

For relaxed support,

$$\bigcup_{\substack{G \text{ connected and noncomplete} \\ |V(G)|=n}} \mathcal{E}_{\geq 0}(G) = \left\{ 0 \leq \lambda_2 \leq \cdots \leq \lambda_n : \lambda_n \geq \frac{n}{n-2}\lambda_2 \right\}, \tag{31}$$

and the single graph $K_n - e$ realizes the entire right-hand side.

Proof. Theorem 3.1 gives the inclusion from left to right. For $n \geq 5$, Theorem 1.2 shows that $K_n - e$ realizes the whole right-hand side. For $n = 4$, Theorem 1.3 does the same except on $(t, t, 2t)$, which is realized by the uniformly weighted cycle C_4. For $n = 3$, Theorem 1.4 gives the exact path condition $\lambda_3 \geq 3\lambda_2$. The relaxed assertion is Theorem 1.1. □

Without a support constraint, positive spectra of symmetric doubly stochastic matrices are classical, by Martignon [13]. Graph-constrained symmetric nonnegative inverse-eigenvalue problems also have an established literature, including Leal-Duarte and Johnson [12]. The next result is the exact positive-semidefinite symmetric doubly stochastic inverse-eigenvalue (PSD–SDIEP) slice with off-diagonal support exactly $K_n - e$.

Corollary 6.4 (An exact $K_n - e$-support PSD–SDIEP slice)**.** *Let $n \geq 3$ and*

$$1 > \nu_2 \geq \cdots \geq \nu_n = a \geq 0.$$

There exists an entrywise nonnegative, positive semidefinite, symmetric doubly stochastic matrix A with spectrum $(1, \nu_2, \dots, \nu_n)$ and off-diagonal support exactly $K_n - e$ if and only if

$$\nu_2 \geq a + \frac{2(1-a)}{n} = \frac{2 + (n-2)a}{n}, \tag{32}$$

and, when $n = 4$, one does not have

$$\nu_2 = \nu_3 = \frac{1+a}{2}. \tag{33}$$

For relaxed support, (32) alone is necessary and sufficient for every $n \geq 3$.

Proof. Since $a < 1$, consider the affine transformation

$$C = \frac{A - aI}{1-a}, \qquad A = aI + (1-a)C$$

and note that $a = \lambda_{\min}(A)$. Thus $A - aI \succeq 0$, so its diagonal entries are nonnegative, while its off-diagonal entries are unchanged up to the positive factor $(1-a)^{-1}$. It follows that C is entrywise nonnegative, positive semidefinite, symmetric and doubly stochastic, with the same off-diagonal support as A. Its spectrum is

$$\left(1, \frac{\nu_2 - a}{1-a}, \dots, \frac{\nu_{n-1} - a}{1-a}, 0\right).$$

Now $L = I - C$ is a weighted Laplacian with largest eigenvalue one. Conversely, a realizing Laplacian normalized by $\lambda_n(L) = 1$ satisfies $0 \preceq L \preceq I$, so $C = I - L$ is positive semidefinite and entrywise nonnegative: its off-diagonal entries are the edge weights and its diagonal entries are nonnegative by positive semidefiniteness. Moreover $C\mathbf{1} = \mathbf{1}$, and $A = aI + (1-a)C$ has all the required properties. Thus the correspondence is reversible. The missing-edge condition becomes

$$1 \geq \frac{n}{n-2}\left(1 - \frac{\nu_2 - a}{1-a}\right),$$

which is (32). Apply Theorems 1.1, 1.2, and 1.3, together with the $n = 3$ case of Theorem 1.4. The unique order-four closure-only spectrum is characterized by the first two positive eigenvalues of L both being $1/2$, which transforms exactly into (33). □

Corollary 6.5 (The Fallat–Gupta–Lin boundary question)**.** *Within the ordered chamber $0 \leq x \leq y \leq z$, the closed realizable cones are*

$$\begin{aligned} \mathcal{E}_{\geq 0}(\text{Paw}) &= \{(x,y,z) : z \geq (2+\sqrt{3})x\}, \\ \mathcal{E}_{\geq 0}(C_4) &= \{(x,y,z) : z \geq x + y\}, \\ \mathcal{E}_{\geq 0}(K_4 - e) &= \{(x,y,z) : z \geq 2x\}. \end{aligned}$$

Consequently, in the relative interior of the ordered chamber their nontrivial graph-dependent boundary facets are, respectively,

$$z = (2+\sqrt{3})x, \qquad z = x + y, \qquad z = 2x.$$

Under the normalization $x + y + z = 2|E(G)|$ used in [6], the corresponding boundary segments are

$$\text{Paw}: \ y = 8 - (3+\sqrt{3})x, \qquad C_4: \ y = 4 - x, \qquad K_4 - e: \ y = 10 - 3x.$$

These are the graph-dependent outer boundary segments of the closed spectral images. Combining the first two descriptions from [5] with the $K_4 - e$ description proved here gives an affirmative answer to [6, Section 5, item 1].

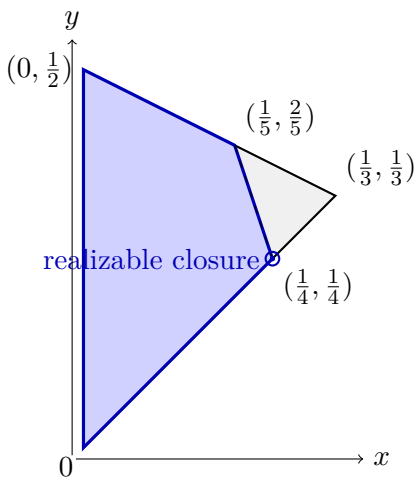


FIGURE 1. The trace-one ordered chamber (gray) and the closed spectral region of $K_4 - e$ (blue). The dashed line is $z = 2x$; at $(1/4, 1/4)$ the filled center records membership in the closure, while the surrounding ring marks the strictly unattained ray $(x, x, 2x)$.

Proof. Erchenko, Jakobson, and Tsypin show that a nonnegative triple is realizable by C_4 exactly when its largest coordinate is at least half its sum [5, Theorem 4.1]. In the ordered chamber this is $z \geq x + y$. Their kite graph is the paw of [6]; its region is described by the condition that at least one pair u, v satisfies $\sqrt{3}|u - v| \geq u + v$ [5, Theorem 6.1]. After ordering, the union of the three pair conditions is equivalent to $\sqrt{3}(z - x) \geq z + x$, or $z \geq (2 + \sqrt{3})x$. The third equality is Theorem 1.3. Its strict-support ray $(t, t, 2t)$ is restored upon taking the closure, so it does not alter the closed boundary facet. □

The geometry becomes especially transparent on a trace slice. Normalize $x + y + z = 1$ and project to the (x, y)-plane. The ordered closed region is

$$x \geq 0, \qquad y \geq x, \qquad x + 2y \leq 1, \qquad 3x + y \leq 1. \tag{34}$$

Its vertices are

$$(0, 0), \quad (0, 1/2), \quad (1/5, 2/5), \quad (1/4, 1/4).$$

The last vertex is the normalized exceptional ray and is not attained with all five weights positive. Figure 1 depicts this trace-one region.

Corollary 6.6 (Exact spectral volume)**.** *The normalized spectral volume of $K_4 - e$ is*

$$P(K_4 - e) = \frac{9}{10}.$$

Proof. The uniform measure on Δ_3 is the distribution of $(E_1, E_2, E_3)/(E_1 + E_2 + E_3)$ for independent unit exponentials. Since the condition is homogeneous, the excluded portion is

$$\Pr(E_{(3)} < 2E_{(1)}).$$

For a specified variable to be the minimum at t, the other two variables must lie in $[t, 2t]$. Hence, up to a null boundary,

$$\Pr(E_{(3)} < 2E_{(1)}) = 3 \int_0^\infty e^{-t}(e^{-t} - e^{-2t})^2 \, dt = \frac{1}{10}.$$

The complement has volume $9/10$. □

Remark 6.7 (A genus-three multiscale boundary chart). A maximally pinched closed genus-three surface decomposes into four pairs of pants, and one possible trivalent dual graph is K_4; cf. [5]. In a two-scale degeneration for which five effective cuff conductances are of order s and the sixth is $o(s)$, the order-s Burger graph Laplacian has positive support $K_4 - e$. Thus $K_4 - e$ occurs as a possible tangent support on the boundary of the K_4 weight simplex. This is only a tangent-graph observation. No exact hyperbolic spectral prescription or maximal-cusp obstruction is asserted or used here.

7. Strict-support spectral-pair constructions

We now prove Theorem 1.4. The construction is explicit except for one use of complete-graph flexibility.

Proof of Theorem 1.4. Label the missing edge 12, put $s = n-2$ and $\rho = n/s$, and let Λ' be the multiset left after deleting the two distinguished occurrences α, β from Λ. Define

$$p = \frac{e_1 - e_2}{\sqrt{2}}, \qquad q = \sqrt{\frac{s}{2n}}(e_1 + e_2) - \sqrt{\frac{2}{ns}} \sum_{j=3}^{n} e_j, \tag{35}$$

and

$$W = \left\{ v \in \mathbb{R}^n : v_1 = v_2 = 0,\ \sum_{j=3}^{n} v_j = 0 \right\}.$$

Then p, q are orthonormal vectors in $\mathbf{1}^\perp$, and

$$\mathbf{1}^\perp = \operatorname{span}\{p, q\} \perp W.$$

Set

$$u = \frac{\alpha + \beta}{1 + \rho}, \qquad v = \rho u, \qquad \eta = \sqrt{(u - \alpha)(\beta - u)}. \tag{36}$$

The assumption $\beta \geq \rho\alpha$ is equivalent to $u \geq \alpha$, while $u \leq \beta$ is automatic. Thus η is real. Moreover,

$$u + v = \alpha + \beta, \qquad uv - \eta^2 = \alpha\beta,$$

so

$$M = \begin{pmatrix} u & \eta \\ \eta & v \end{pmatrix}$$

has eigenvalues α, β.

Put

$$\delta = \frac{2v}{n} = \frac{\alpha + \beta}{n - 1}. \tag{37}$$

Every $t \in \Lambda'$ satisfies $t - \delta > 0$. By complete-graph flexibility [6, Theorem 2.5], there is a strictly weighted Laplacian $R \in \mathcal{S}_L(K_s)$ whose nonzero spectral multiset is

$$\{t - \delta : t \in \Lambda'\}. \tag{38}$$

When $n = 3$, the space W is zero-dimensional and Λ' is empty; take $R = 0$. Let

$$H_s = I_s - \frac{1}{s} J_s$$

and embed $\delta H_s + R$ in the coordinates $3, \ldots, n$. Define

$$L = upp^{\mathsf{T}} + \eta(pq^{\mathsf{T}} + qp^{\mathsf{T}}) + vqq^{\mathsf{T}} + \iota(\delta H_s + R), \tag{39}$$

where ι denotes the central-coordinate embedding.

The summands in (39) preserve

$$\mathbb{R}^n = \operatorname{span}\{\mathbf{1}\} \perp \operatorname{span}\{p, q\} \perp W.$$

The restriction to $\operatorname{span}\{p,q\}$ is M, and the restriction to W is $\delta I + R|_W$. Equations (38) and (36) therefore give

$$\operatorname{Spec}(L) = \{0\} \cup \Lambda.$$

It remains to verify the graph pattern. Since $v = nu/s$,

$$L_{12} = -\frac{u}{2} + \frac{vs}{2n} = 0.$$

For $j \geq 3$, direct use of (35) gives

$$L_{1j} = -\frac{v}{n} - \frac{\eta}{\sqrt{ns}} < 0, \qquad L_{2j} = -\frac{v}{n} + \frac{\eta}{\sqrt{ns}}. \tag{40}$$

Furthermore,

$$\eta^2 = uv - \alpha\beta < uv = \frac{s}{n}v^2,$$

so $\eta/\sqrt{ns} < v/n$ and the second entry in (40) is also strictly negative.

Finally, for $3 \leq j \neq k \leq n$, the vqq^{T} contribution is $2v/(ns)$, whereas the δH_s contribution is $-\delta/s = -2v/(ns)$. They cancel, leaving

$$L_{jk} = R_{jk} < 0.$$

Thus $L_{12} = 0$, every other off-diagonal entry is strictly negative, and $L \in \mathcal{S}_L(K_n - e)$. □

Corollary 7.1 (An ordered strict-support cone)**.** *Let $0 < \lambda_2 \leq \cdots \leq \lambda_n$. If*

$$\lambda_3 \geq \frac{n}{n-2}\lambda_2, \tag{41}$$

then $\{0, \lambda_2, \ldots, \lambda_n\}$ has a strict $K_n - e$ realization. The projective volume of this sufficient cone is $(n-1)/(n+1)$.

Proof. Apply Theorem 1.4 with $(\alpha, \beta) = (\lambda_2, \lambda_3)$. Every remaining eigenvalue is at least β, while $(\alpha+\beta)/(n-1) < \beta$ for $n \geq 4$; the case $n = 3$ has no remaining occurrence. For the volume, use independent unit exponentials. If $T = E_{(1)}$ and $D = E_{(2)} - E_{(1)}$, then T and D are independent exponentials of rates $n-1$ and $n-2$, respectively. Hence

$$\Pr\left(E_{(2)} \geq \frac{n}{n-2}E_{(1)}\right) = \mathbb{E}(e^{-2T}) = \frac{n-1}{n+1}.$$

□

At equality $\beta = \rho_n\alpha$, the insertion construction has $\eta = 0$ and lands on the weak block locus, but all allowed edge weights remain positive. Equality in the condition $t > \delta$ is not included: in this construction it would give R an additional zero eigenvalue.

Proof of Theorem 1.5. Necessity follows from Theorem 3.1. For $n = 3$, the spectral-pair insertion theorem has no remaining occurrence and gives a strict realization exactly when $b \geq 3a$. For $n = 4$, the assertion is Theorem 1.3 applied to (a, a, b).

Suppose now that $n \geq 5$, put $r = b/a$, and assume $r \geq \rho_n$. If $r < n-2$, choose $(\alpha, \beta) = (a, b)$ in Theorem 1.4. Every remaining occurrence is $t = a$, and

$$a > \frac{a+b}{n-1} \iff r < n-2,$$

so the resulting support is strict.

It remains to treat $r \geq n-2$. Put

$$c = \frac{a}{n(b-a)} = \frac{1}{n(r-1)}, \qquad h = \sqrt{\frac{1+2c}{2}}.$$

Since $n \geq 5$ and $r \geq n-2$, one has $0 < c < 1/6$. Let x, y be the two positive roots of

$$z^2 - hz + c = 0$$

and set

$$q = (x, y, -h, 0, \ldots, 0)^{\mathsf{T}}.$$

The discriminant is $(1-6c)/2 > 0$, and

$$q^{\mathsf{T}}\mathbf{1} = x + y - h = 0, \qquad \|q\|^2 = x^2 + y^2 + h^2 = 2h^2 - 2c = 1, \qquad q_1 q_2 = c.$$

Every other product $q_i q_j$ is either negative or zero, and hence is strictly smaller than c. Therefore

$$L = aH_n + (b-a)qq^{\mathsf{T}}$$

has spectrum $\{0, a^{(n-2)}, b\}$ and satisfies

$$L_{12} = -\frac{a}{n} + (b-a)c = 0, \qquad L_{ij} = -\frac{a}{n} + (b-a)q_i q_j < 0 \quad (\{i,j\} \neq \{1,2\}).$$

Thus $L \in \mathcal{S}_L(K_n - e)$, completing the classification. □

8. Two disjoint missing edges: orbitopes and an exact rank-two slice

The first higher-defect case already separates linear spectral obstructions from the nonlinear support-incidence problem. We begin with the general linear family. Write $H_n = I_n - J_n/n$ as in Section 2.

Theorem 8.1 (Complement-orbitope inequalities)**.** *Let G be a graph on n vertices, put $F = \overline{G}$, and let W be any symmetric zero-diagonal matrix supported on $E(F)$. Write*

$$\theta_1(W) \leq \cdots \leq \theta_{n-1}(W)$$

for the eigenvalues of $H_n W H_n$ on $\mathbf{1}^{\perp}$. If the ordered list

$$0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$$

is realized by a relaxed weighted Laplacian on G, then

$$\sum_{i=1}^{n-1} \lambda_{i+1}\theta_i(W) \geq 0. \tag{42}$$

Proof. The claim is immediate for the zero spectrum. Otherwise put

$$B = I - \frac{1}{\lambda_n}L = \frac{1}{n}J_n + C.$$

On $\mathbf{1}^{\perp}$ the eigenvalues of C in decreasing order are $1 - \lambda_i/\lambda_n$, $2 \leq i \leq n$. For every nonedge ij one has $C_{ij} = -1/n$. Since W is supported on the nonedges,

$$\operatorname{tr}(CW) = -\frac{2}{n}\sum_{ij \in E(F)} W_{ij} = \operatorname{tr}(H_n W H_n).$$

Moreover $C = H_n C H_n$, and hence

$$\operatorname{tr}(CW) = \operatorname{tr}\big(C(H_n W H_n)\big).$$

The von Neumann trace inequality on $\mathbf{1}^{\perp}$ [1, Chapters II–III], pairing the decreasing eigenvalues of C with the increasing $\theta_i(W)$, gives

$$\operatorname{tr}(CW) \geq \sum_{i=1}^{n-1}\left(1 - \frac{\lambda_{i+1}}{\lambda_n}\right)\theta_i(W).$$

The left side equals $\sum_i \theta_i(W)$. Cancelling the trace terms and multiplying by λ_n proves (42). □

Now take $F = M_2$ with missing edges 12 and 34, and put

$$h = \frac{n-4}{n}. \tag{43}$$

The symmetric choice in Theorem 8.1 gives the facet

$$\lambda_n + h\lambda_{n-1} \geq \lambda_2 + \lambda_3. \tag{44}$$

Allowing arbitrary real coefficients on the two missing edges produces no additional linear information.

Theorem 8.2 (Collapse of the M_2 complement orbitope)**.** *Let $n \geq 5$. For $x, y \in \mathbb{R}$, let $W(x,y)$ be the symmetric zero-diagonal matrix whose only possibly nonzero upper-triangular entries are $W_{12} = x$ and $W_{34} = y$. Every nonnegative ordered list satisfying (44) also satisfies*

$$\sum_{i=1}^{n-1} \lambda_{i+1}\theta_i(W(x,y)) \geq 0 \qquad (x, y \in \mathbb{R}).$$

Conversely, the inequality for $x = y = 1$ is exactly (44). Thus, in the ordered chamber, the complete family of complement-orbitope inequalities supported on M_2 is equivalent to the single symmetric facet (44).

Proof. Put $m = n-1$, $\ell = (\lambda_2, \dots, \lambda_n)$, and

$$v_j = (0^{j-1}, 1^{m-j+1}) \qquad (1 \leq j \leq m).$$

Every nonnegative ordered list has a unique expansion $\ell = \sum_{j=1}^{m} d_j v_j$ with $d_j \geq 0$. Define

$$\mathcal{F}(\ell) = \ell_1 + \ell_2 - h\ell_{m-1} - \ell_m.$$

Then (44) is $\mathcal{F}(\ell) \leq 0$, and

$$\mathcal{F}(v_1) = 1 - h, \qquad \mathcal{F}(v_2) = -h,$$
$$\mathcal{F}(v_j) = -(1+h) \quad (3 \leq j \leq m-1), \qquad \mathcal{F}(v_m) = -1.$$

There is exactly one positive coefficient. In the d-coordinates, the extreme rays of $d_j \geq 0$, $\mathcal{F}(\ell) \leq 0$ are therefore

$$v_j \quad (2 \leq j \leq m), \qquad -\mathcal{F}(v_j)v_1 + (1-h)v_j \quad (2 \leq j \leq m).$$

Let $\theta_1 \leq \cdots \leq \theta_m$ be the spectrum of $H_n W(x,y) H_n$ on $\mathbf{1}^{\perp}$, and put $T_j = \sum_{i=j}^{m} \theta_i$. Testing $\theta \cdot \ell \geq 0$ on the displayed extreme rays reduces the assertion to

$$T_j \geq 0 \qquad (2 \leq j \leq m), \tag{45}$$
$$hT_1 + (1-h)T_2 \geq 0, \tag{46}$$
$$(1+h)T_1 + (1-h)T_j \geq 0 \qquad (3 \leq j \leq m-1), \tag{47}$$
$$T_1 + (1-h)T_m \geq 0. \tag{48}$$

The edge-difference vectors $e_1 - e_2$ and $e_3 - e_4$ are eigenvectors of the compression with eigenvalues $-x$ and $-y$. The subspace supported on vertices $5, \dots, n$ with coordinate sum zero contributes 0^{n-5}. On the remaining two-dimensional invariant quotient, a direct compression in the two pair-sum directions has trace $(n-2)(x+y)/n$ and determinant $(n-4)xy/n$. Hence the characteristic polynomial is

$$z^{n-5}(z+x)(z+y)\left(z^2 - \frac{(n-2)(x+y)}{n}z + \frac{(n-4)xy}{n}\right). \tag{49}$$

Suppose first that $x \geq y \geq 0$. If $r_- \leq r_+$ are the two quadratic roots in (49), the ordered spectrum is

$$-x, -y, 0^{n-5}, r_-, r_+.$$

Consequently

$$T_1 = -\frac{2(x+y)}{n}, \qquad T_2 = x - \frac{2(x+y)}{n}, \qquad T_j = \frac{n-2}{n}(x+y) \quad (3 \le j \le m-1).$$

These quantities give (45); the left sides of (46) and (47) reduce respectively to $2(x-y)/n$ and 0. Evaluating the quadratic factor at $(x+y)/2$ gives

$$\frac{n-4}{n}\left(xy - \frac{(x+y)^2}{4}\right) \le 0,$$

so $r_+ \ge (x+y)/2$ and (48) follows. If $x, y \le 0$, negating the preceding spectrum shows directly that T_1 and all the tails occurring in (45)–(48) are nonnegative.

It remains to take $x = a > 0$ and $y = -b < 0$. Write the two quadratic roots as $-u < 0 < v$. They satisfy

$$v - u = \frac{n-2}{n}(a-b), \qquad uv = \frac{n-4}{n}ab. \tag{50}$$

The negative eigenvalues are $-a, -u$, the positive eigenvalues are b, v, and all remaining eigenvalues are zero. Put $A = \max(a, u)$ and $B = \max(b, v)$. Then

$$T_1 = \frac{2(b-a)}{n}, \qquad T_2 = T_1 + A,$$

while every tail in (47) is $b + v$. This proves (45). The left side of (46) is

$$\frac{2(b-a) + 4A}{n} > 0.$$

If $b \ge a$, (47) and (48) are immediate. If $a > b$, multiplying the left side of (47) by $n/4$ and using (50) gives

$$b + v - \frac{n-2}{n}(a-b) = b + u > 0.$$

Also $B \ge v \ge (n-2)(a-b)/n > (a-b)/2$, which proves (48). Swapping x and y leaves the spectrum unchanged, so all real cases have been covered.

For $x = y = 1$, (49) gives

$$-1, -1, 0^{n-5}, h, 1,$$

and the orbitope inequality is precisely (44). □

The collapse theorem is a limitation theorem: varying a linear witness on the two missing edges cannot prove more than (44). The full $K_n - M_2$ cone still requires a nonlinear fixed-spectrum orbit-incidence argument. The following slice can nevertheless be closed exactly.

Theorem 8.3 (Exact two-level top-rank-two slice)**.** *Let $n \ge 6$ and $0 < a < b$. Put $g = b/a - 1$. The spectrum*

$$\{0, a^{(n-3)}, b^{(2)}\} \tag{51}$$

has a relaxed $K_n - M_2$ realization if and only if

$$g \ge \frac{2}{n-2}.$$

It has a strict $K_n - M_2$ realization if and only if

$$\begin{cases} g \ge 2/(n-2), & n \ge 7, \\ g > 1/2, & n = 6. \end{cases}$$

Thus the unique boundary exception in this range occurs for $n = 6$ and $b/a = 3/2$.

Proof. Put

$$c = \frac{1}{ng}.$$

A rank-two orthogonal projection P with $P\mathbf{1} = 0$ gives

$$L = aH_n + agP.$$

This matrix has spectrum (51), and, for $i \neq j$,

$$L_{ij} = ag(P_{ij} - c).$$

Hence it has exact strict $K_n - M_2$ support if and only if

$$P_{12} = P_{34} = c, \qquad P_{ij} < c \quad (ij \neq 12, 34). \tag{52}$$

For relaxed support, the final strict inequalities are replaced by $P_{ij} \leq c$.

Substitution in (44) gives the necessary condition $g \geq 2/(n-2)$. At $n = 6$ and $g = 1/2$ one has $c = 1/3$. If A_F is the adjacency matrix of the matching, then

$$\mathrm{Spec}(H_6 A_F H_6|_{\mathbf{1}^\perp}) = (-1, -1, 0, 1/3, 1).$$

For any rank-two projection satisfying the two contacts,

$$\mathrm{tr}(PH_6A_FH_6) = 2(P_{12} + P_{34}) = \frac{4}{3}.$$

This is the rank-two Ky Fan maximum $1 + 1/3$ [1, Chapter III]. Equality forces $\mathrm{im}\, P$ to be the sum of the two positive eigenspaces and therefore forces

$$P = P_0 := \frac{1}{6}\begin{pmatrix} 2 & 2 & -1 & -1 & -1 & -1 \\ 2 & 2 & -1 & -1 & -1 & -1 \\ -1 & -1 & 2 & 2 & -1 & -1 \\ -1 & -1 & 2 & 2 & -1 & -1 \\ -1 & -1 & -1 & -1 & 2 & 2 \\ -1 & -1 & -1 & -1 & 2 & 2 \end{pmatrix}.$$

In particular $(P_0)_{56} = c$, so a third zero edge is forced and exact strict support is impossible at this boundary point. The same P_0 does give a relaxed realization.

It remains to construct projections satisfying (52). Put $q = n - 4$ and define

$$A = \frac{1}{2}(1, 1, -1, -1, 0^q), \qquad B = (u, u, u, u, v, \ldots, v),$$

where

$$u = \sqrt{\frac{q}{4n}}, \qquad v = -\frac{2}{\sqrt{nq}}.$$

Also set

$$E = \frac{1}{\sqrt{2}}(0, 0, 0, 0, 1, -1, 0^{q-2}).$$

These vectors lie in $\mathbf{1}^\perp$ and A, B, E are mutually orthonormal. For $0 \leq s \leq \pi/2$ let

$$B_s = (\cos s)B + (\sin s)E, \qquad P_s = AA^{\mathsf{T}} + B_s B_s^{\mathsf{T}}.$$

Then

$$(P_s)_{12} = (P_s)_{34} = \frac{1}{4} + \frac{q}{4n}\cos^2 s, \tag{53}$$

so this family covers $1/4 \leq c \leq (n-2)/(2n)$.

For completeness, we verify strictness. Cross-pair entries among the first four vertices equal $c - 1/2$. Every endpoint–core entry is at most $u/(2\sqrt{2}) < 1/4 \le c$. The only potentially largest core–core entry is

$$v^2 \cos^2 s + \frac{|v|}{\sqrt{2}} \cos s \sin s.$$

Writing $d = u^2 - v^2$, its difference from c has minimum

$$\frac{1}{4} + \frac{d}{2} - \frac{1}{2}\sqrt{d^2 + \frac{v^2}{2}}.$$

Here $1/2 + d = (3q-4)/(4q) > 0$ for $q \ge 2$, so squaring introduces no change of sign. Positivity is equivalent to

$$\frac{1}{4} + u^2 - \frac{3}{2}v^2 > 0,$$

or, after multiplication by $4q(q+4)$, $2(q+6)(q-2) > 0$. Hence every other entry is strictly below c for $n \ge 7$, including at $s = 0$. For $n = 6$ the same conclusion holds for $s > 0$, while

$$c - (P_s)_{56} = \frac{3}{4}\sin^2 s;$$

the sole failure is $s = 0$. Thus (53) covers the required range down to $c = 1/4$.

For $0 < c \le 1/4$, define

$$C = \frac{1}{2}(1, -1, 1, -1, 0^{n-4})$$

and, for $0 \le t < \pi/4$, put

$$A_t = (\cos t)A + (\sin t)C, \qquad Q_t = A_t A_t^{\mathsf{T}} + EE^{\mathsf{T}}.$$

Then

$$(Q_t)_{12} = (Q_t)_{34} = \frac{\cos 2t}{4}.$$

The other entries among the first four vertices are

$$-c, \quad -\frac{(\cos t + \sin t)^2}{4}, \quad -\frac{(\cos t - \sin t)^2}{4}, \quad -c;$$

all endpoint–core entries are zero, $(Q_t)_{56} = -1/2$, and every remaining core entry is zero. They are all strictly below c.

The two constructions therefore prove sufficiency for $n \ge 7$ when $c \le (n-2)/(2n)$, equivalently $g \ge 2/(n-2)$, and for $n = 6$ when $c < 1/3$, equivalently $g > 1/2$. Necessity and the order-six exception were proved above; P_0 supplies the missing relaxed order-six endpoint. □

Remark 8.4 (Scope of the slice). Theorem 8.3 does not treat $n = 5$, other intermediate eigenspace ranks, or arbitrary spectra, and it does not assert that (44) is sufficient for the full $K_n - M_2$ cone. The repeated-top slice of rank $n - 2$ has a separate exact-support obstruction.

9. Exact spectral volume and the last-edge gap

The half-space description in Theorem 1.1 makes the spectral volume exactly computable.

Theorem 9.1 (Exact spectral volume of $K_n - e$)**.** *For every $n \ge 3$,*

$$P(K_n - e) = 1 - \binom{(n-2)(n+1)/2}{n-2}^{-1}. \tag{54}$$

Consequently $P(K_n - e) \to 1$.

Proof. Put $m = n - 1$. A uniformly distributed point of Δ_m is represented by

$$\left(\frac{E_1}{\sum_i E_i}, \dots, \frac{E_m}{\sum_i E_i}\right),$$

where $E_1, \dots, E_m$ are independent unit exponentials. We use the classical exponential representation and order-statistic spacings [14]. By Theorem 1.1, the excluded event is

$$E_{(m)} < \left(1 + \frac{2}{m-1}\right) E_{(1)}.$$

With $c = 2/(m-1)$, conditioning on which variable is minimal and on its value gives

$$\begin{aligned}\Pr(E_{(m)} < (1+c)E_{(1)}) &= m \int_0^\infty e^{-mt}(1 - e^{-ct})^{m-1}\, dt \\ &= \frac{m}{c} B\left(\frac{m}{c}, m\right) \\ &= \binom{(n-2)(n+1)/2}{n-2}^{-1}.\end{aligned}$$

Here $m/c = m(m-1)/2$ and $m/c + m - 1 = (n-2)(n+1)/2$. Taking the complement proves the formula. □

Corollary 9.2 (The exact last-edge gap)**.** *For every $n \geq 3$,*

$$P(K_n) - P(K_n - e) = \binom{(n-2)(n+1)/2}{n-2}^{-1}. \tag{55}$$

Writing $k = n - 2$ and $N = k(k+3)/2$, one has

$$1 - P(K_n - e) \sim e^{-2}\sqrt{2\pi k}\left(\frac{2}{ek}\right)^k. \tag{56}$$

Moreover,

$$P(K_{n+1} - e) > P(K_n - e),$$

so one-step suspension monotonicity holds on the entire one-missing-edge family.

Proof. The complete-graph theorem gives $P(K_n) = 1$, proving (55). Write

$$N^{\underline{k}} = N(N-1)\cdots(N-k+1).$$

Then

$$1 - P(K_n - e) = \frac{k!}{N^{\underline{k}}}, \qquad \frac{N^{\underline{k}}}{N^k} = \prod_{j=0}^{k-1}\left(1 - \frac{j}{N}\right) \longrightarrow e^{-1}.$$

Together with Stirling's formula and $N^k \sim e^3(k^2/2)^k$, this gives (56).

Finally, the binomial coefficient in (55) increases strictly with n. Indeed,

$$\binom{N+k+2}{k+1} \geq \binom{N+1}{k+1} = \frac{N+1}{k+1}\binom{N}{k} > \binom{N}{k}.$$

Thus its reciprocal decreases strictly. □

Remark 9.3. This does not decide whether there is a positive edge-gain lower bound uniform over all connected noncomplete graphs of a fixed order n. It does show that any such constant satisfies

$$c_n \leq \binom{(n-2)(n+1)/2}{n-2}^{-1}.$$

If disconnected graphs are admitted literally, zero-gap examples already make an unrestricted version degenerate.

10. Fixed-core joins and iterated suspension

For graphs on disjoint vertex sets, write $G \vee H$ for their join: all edges of G and H are retained and every vertex of G is joined to every vertex of H. Thus the k-fold suspension of G is $G \vee K_k$. The one-vertex suspension formula goes back at least to Colin de Verdière [4]; related Laplacian formulas for generalized graph compositions appear in [9], and the block formula below is the weighted join construction of [6, Lemma 2.4]. We do not claim the join identity itself as new; the new step is its probabilistic use to control spectral volume.

We first record the local input that allows a fixed graph to absorb a prescribed finite portion of a large spectrum.

Lemma 10.1 (A spectral box)**.** *Let H be a connected graph on $m \geq 2$ vertices. There are bounded open intervals*

$$I_1 < I_2 < \cdots < I_{m-1} \subset \mathbb{R}_{>0}$$

such that every tuple

$$(\mu_1, \ldots, \mu_{m-1}) \in I_1 \times \cdots \times I_{m-1}$$

is the nonzero spectrum of some matrix in $\mathcal{S}_L(H)$.

Proof. By [3, Theorem 3.11], the class $\mathcal{S}_L(H)$ contains a matrix A with the SSPWL. Choose a real symmetric matrix M with $M\mathbf{1}_m = 0$ and simple spectrum, arbitrarily close to A. The Bifurcation lemma [3, Lemma 2.12] gives $A_0 \in \mathcal{S}_L(H)$ with $\mathrm{Spec}(A_0) = \mathrm{Spec}(M)$ and the SSPWL. Thus

$$\mathrm{Spec}(A_0) = \{0, \mu_1^*, \ldots, \mu_{m-1}^*\}, \qquad 0 < \mu_1^* < \cdots < \mu_{m-1}^*.$$

Choose an orthonormal eigenbasis of A_0 whose first vector is $m^{-1/2}\mathbf{1}_m$. Replacing the nonzero diagonal entries in this basis by any sufficiently close tuple $(\mu_1, \ldots, \mu_{m-1})$ produces a matrix M' with $M'\mathbf{1}_m = 0$ arbitrarily close to A_0. Another application of the Bifurcation lemma gives a matrix in $\mathcal{S}_L(H)$ with spectrum $\mathrm{Spec}(M')$. Taking sufficiently small pairwise disjoint intervals around the μ_i^* proves the claim. This is also the local spectral openness encoded by the full-rank Jacobian criterion in [3, Theorem 3.8]. □

Lemma 10.2 (Join spectrum)**.** *Let H have m vertices, let $k \geq 1$, and put $N = m + k$. Suppose*

$$\mathrm{Spec}(A) = \{0, \mu_1, \ldots, \mu_{m-1}\}, \qquad A \in \mathcal{S}_L(H),$$

and

$$\mathrm{Spec}(B) = \{0, \nu_1, \ldots, \nu_{k-1}\}, \qquad B \in \mathcal{S}_L(K_k),$$

where for $k = 1$ the second list is empty and $B = (0)$. For $c > 0$, set

$$L(A, B, c) = \begin{pmatrix} A + kcI_m & -cJ_{m,k} \\ -cJ_{k,m} & B + mcI_k \end{pmatrix}. \tag{57}$$

Then $L(A, B, c) \in \mathcal{S}_L(H \vee K_k)$ and

$$\mathrm{Spec}\, L(A, B, c) = \{0, Nc\} \cup \{\mu_i + kc : 1 \leq i \leq m-1\} \cup \{\nu_j + mc : 1 \leq j \leq k-1\}. \tag{58}$$

Proof. The matrix in (57) retains the weights of A and B and assigns weight c to every joining edge. The four mutually orthogonal spaces

$$\mathrm{span}\{\mathbf{1}_N\}, \quad \{(u, 0) : u \perp \mathbf{1}_m\}, \quad \{(0, v) : v \perp \mathbf{1}_k\}, \quad \mathrm{span}\{(k\mathbf{1}_m, -m\mathbf{1}_k)\}$$

are invariant. The corresponding eigenvalues are, respectively, 0, the numbers $\mu_i + kc$, the numbers $\nu_j + mc$, and Nc, proving (58). □

We now combine this deterministic splitting with exponential order statistics. The quantitative estimate will also verify that all edge weights may be kept strictly positive.

Theorem 10.3 (Fixed-core join theorem)**.** *Let H be a fixed connected graph on $m \geq 2$ vertices. There are constants $\eta > 0$ and $p_1, \dots, p_{m-1} \in (0,1)$, depending only on H, such that for every $k \geq 1$, with $N = m + k$,*

$$P(H \vee K_k) \geq 1 - \exp\left(-\frac{\eta N(N-1)}{m}\right) - \sum_{i=1}^{m-1} (1-p_i)^{N-2}. \tag{59}$$

In particular, there are $C_H, c_H > 0$ such that

$$1 - P(H \vee K_k) \leq C_H e^{-c_H k}. \tag{60}$$

Every spectrum counted in the lower bound has a realization with strictly positive weight on every edge of $H \vee K_k$.

Proof. Choose the spectral box $I_1 \times \cdots \times I_{m-1}$ from Lemma 10.1. Choose nonempty open intervals $J_i \Subset I_i$ and $\eta > 0$ such that

$$J_i + [0, \eta] \subset I_i \qquad (1 \leq i \leq m-1). \tag{61}$$

For a unit exponential random variable Y, put

$$p_i = \Pr(Y \in J_i) > 0.$$

Let $X_1, \dots, X_{N-1}$ be independent unit exponential random variables. Their normalization by $\sum_j X_j$ is uniformly distributed on the trace-one simplex. Since spectral realizability is homogeneous, it suffices to realize the unnormalized multiset $\{X_1, \dots, X_{N-1}\}$.

Let $T = \min_j X_j$. Almost surely the minimum is unique. Conditional on its minimizing index, the exponential memoryless property says that the $N-2$ residual differences

$$Y_\ell = X_\ell - T$$

are independent unit exponentials, independent of T, while T is exponential with rate $N-1$. Consider the event

$$\mathcal{A} = \left\{ \frac{mT}{N} \leq \eta \right\} \cap \bigcap_{i=1}^{m-1} \{\text{some } Y_\ell \text{ belongs to } J_i\}. \tag{62}$$

The intervals J_i are disjoint, so on $\mathcal{A}$ the selected indices may be chosen distinctly. A union bound gives

$$\begin{aligned} \Pr(\mathcal{A}^c) &\leq \Pr\left(T > \frac{\eta N}{m}\right) + \sum_{i=1}^{m-1} \Pr\left(Y_\ell \notin J_i \text{ for every } \ell\right) \\ &= \exp\left(-\frac{\eta N(N-1)}{m}\right) + \sum_{i=1}^{m-1} (1-p_i)^{N-2}. \end{aligned} \tag{63}$$

Suppose $\mathcal{A}$ occurs and put

$$c = \frac{T}{N}, \qquad \mu_i = Y_{\ell_i} + \frac{mT}{N},$$

where $Y_{\ell_i} \in J_i$. By (61), the tuple $(\mu_1, \dots, \mu_{m-1})$ lies in the spectral box, so it is realized by some $A \in \mathcal{S}_L(H)$. For every unselected residual Y_ℓ, set

$$\nu_\ell = T + Y_\ell - mc = Y_\ell + \frac{kT}{N} > 0.$$

There are exactly $k-1$ such numbers. Complete-graph flexibility [6, Theorem 2.5] provides $B \in \mathcal{S}_L(K_k)$ having them as its nonzero spectrum; for $k = 1$ this list is empty and $B = (0)$.

Apply Lemma 10.2. Its distinguished eigenvalue is $Nc = T$. For each selected index,

$$\mu_i + kc = Y_{\ell_i} + \frac{mT}{N} + \frac{kT}{N} = T + Y_{\ell_i},$$

and for every unselected index,

$$\nu_\ell + mc = T + Y_\ell.$$

Thus $L(A, B, c)$ realizes the entire target multiset. Since A and B have strict support and $c = T/N > 0$, the realization has strict support on $H \vee K_k$. Equation (63) proves (59). The right side is $O_H(e^{-c_H k})$ for a positive c_H, proving (60). □

Proof of Theorem 1.6. Associativity of graph join gives $\mathrm{susp}^k(G) = G \vee K_k$. If G is connected and has at least two vertices, the conclusion is Theorem 10.3. If G is disconnected, then $H = G \vee K_1$ is connected and, for $k \geq 2$,

$$G \vee K_k = H \vee K_{k-1}.$$

Apply Theorem 10.3 to H and absorb the index shift into the constant; increasing that constant also covers the single value $k = 1$. If $G = K_1$, then $G \vee K_k = K_{k+1}$ and $P(G \vee K_k) = 1$. Thus [5, Question 5] has an affirmative answer, with exponential convergence. □

Theorem 10.4 (Uniform finite-defect flexibility)**.** *For every integer $r \geq 0$ there are constants $C_r, c_r > 0$ such that, for every $n \geq 2$,*

$$\sup_{\substack{|V(G)|=n \\ |E(\overline{G})|\leq r}} (1 - P(G)) \leq C_r e^{-c_r n}. \tag{64}$$

Consequently,

$$\lim_{n\to\infty} \inf_{\substack{|V(G)|=n \\ |E(\overline{G})|\leq r}} P(G) = 1.$$

Proof. The case $r = 0$ is the complete-graph theorem. Suppose $r \geq 1$ and first take $n \geq 2r + 2$. If $G = K_n$ there is nothing to prove. Otherwise let S be the set of endpoints of the edges of $\overline{G}$. Then $2 \leq |S| \leq 2r$. Choose $u \in V(G) \setminus S$ and put

$$H = G[S \cup \{u\}].$$

The vertex u is adjacent to every vertex of S, so H is connected. Every vertex outside S is universal in G, and hence

$$G = H \vee K_{n-|V(H)|}, \qquad |V(H)| \leq 2r + 1.$$

For fixed r, only finitely many isomorphism types of such connected cores H occur. Apply Theorem 10.3 to each type and take the minimum of their positive rates and the maximum of their constants. This gives (64) for $n \geq 2r + 2$. There are only finitely many smaller orders, and increasing C_r absorbs them because $0 \leq P(G) \leq 1$. □

11. Tail-admissible cores and the optimal defect exponent

The exponential estimate in Theorem 10.3 uses one fixed spectral box. For several small cores the exact region contains a global cone on which the largest eigenvalue is allowed to move. Overlapping the corresponding join charts improves the error to the optimal scale $\exp\{-n \log n + O(n)\}$.

Definition 11.1 (Tail-admissible core)**.** Let H be a graph on $m \geq 3$ vertices. For $1 \leq \ell \leq m - 2$ and $A > 0$, put

$$\mathcal{T}_{A,\ell}^{(m)} = \left\{ 0 < \mu_1 \leq \cdots \leq \mu_{m-1} : \mu_{m-1} > A \sum_{i=1}^{\ell} \mu_i \right\}.$$

We call H *relaxedly* (A,ℓ)*-tail-admissible* if this cone is contained in $\mathcal{E}_{\geq 0}(H)$, and *strictly* (A,ℓ)*-tail-admissible* if it is contained in $\mathcal{E}_{>0}(H)$. The middle coordinates are unrestricted apart from ordering. The argument below requires $A\ell > 1$.

We first isolate the probabilistic estimate used by the moving-anchor construction. Order statistics are denoted by parentheses; ties have probability zero.

Lemma 11.2 (Exponential clustering)**.** *Let $T \sim \mathrm{Exp}(n-1)$ and let $Y_1, \dots, Y_d$ be independent unit exponentials, independent of T, where $d = n-2$. Suppose that for fixed constants $C, K > 0$,*

$$0 \leq d - r \leq K\frac{n}{\log n}, \qquad 0 < \delta \leq \frac{C}{n}.$$

Then

$$\Pr\Big(Y_{(d)} - Y_{(d-r+1)} \leq \delta\big(Y_{(d)} + T\big)\Big) \leq \exp\{-n\log n + O_{C,K}(n)\}. \tag{65}$$

Proof. Condition on $T = t$, and write $a = \delta t$, $Z = Y_{(d)}$, and $w(z) = \delta z + a$. If $Z \leq 2a$, then

$$\Pr(Z \leq 2a \mid T = t) = (1 - e^{-2a})^d \leq (2a)^d.$$

Averaging over T bounds this contribution by

$$(2\delta)^d \mathbb{E}(T^d) = (2\delta)^d \frac{d!}{(n-1)^d} \leq \exp\{-n\log n + O_{C,K}(n)\}. \tag{66}$$

For $z > 2a$ and all sufficiently large n, one has $w(z) < z$. The exponential mass of $[z - w(z), z]$ is at most

$$p(z,a) = e^{-z+w(z)} - e^{-z} \leq e^{-z} w(z) e^{w(z)}.$$

Choose the coordinate attaining the maximum and the other $r-1$ coordinates in the cluster. Leaving the remaining coordinates unrestricted gives

$$\begin{aligned} &d\binom{d-1}{r-1}\int_{2a}^{\infty} e^{-z} p(z,a)^{r-1}\,dz \\ &\quad\leq d\binom{d-1}{r-1} e^{(r-1)a}\int_0^{\infty} e^{-\beta z}(\delta z + a)^{r-1}\,dz, \end{aligned} \tag{67}$$

where $\beta = r - (r-1)\delta$. The integral is at most

$$2^{r-2}\left(\frac{\delta^{r-1}\Gamma(r)}{\beta^r} + \frac{a^{r-1}}{\beta}\right). \tag{68}$$

Put $\lambda = n - 1$. Since $a = \delta T$,

$$\mathbb{E}e^{(r-1)a} = \frac{\lambda}{\lambda - (r-1)\delta},$$
$$\mathbb{E}\Big(e^{(r-1)a} a^{r-1}\Big) = \frac{\delta^{r-1}\lambda\Gamma(r)}{(\lambda - (r-1)\delta)^r}.$$

Here $\beta \geq r/2$ and $\lambda - (r-1)\delta \geq \lambda/2$ for large n. Thus the average of (67) is at most

$$\delta^{r-1}\exp\{O_{C,K}(n)\} \leq \exp\{-n\log n + O_{C,K}(n)\},$$

because $\log\binom{d-1}{r-1} = O_K(n)$, $\Gamma(r)/(r/2)^r = \exp\{O(n)\}$, and $r = n - O_K(n/\log n)$. Combining this with (66) proves the lemma. □

Theorem 11.3 (Moving-anchor join theorem)**.** *Let H be a fixed connected noncomplete graph on $m \geq 3$ vertices. Suppose that H is relaxedly (A, ℓ)-tail-admissible for some $1 \leq \ell \leq m-2$ with $A\ell > 1$. Then there are $C_H > 0$ and n_H such that, for $n \geq n_H$,*

$$\binom{(n-2)(n+1)/2}{n-2}^{-1} \leq 1 - P(H \vee K_{n-m}) \leq \exp\{-n \log n + C_H n\}. \tag{69}$$

In particular,

$$1 - P(H \vee K_{n-m}) = \exp\{-n \log n + O_H(n)\}. \tag{70}$$

If H is strictly (A, ℓ)-tail-admissible, every spectrum in the construction is realized with strict support. Under relaxed tail-admissibility, the same volume conclusion holds; the possible strict/relaxed discrepancy is null.

Proof. Set

$$M = \left\lfloor \frac{n}{2} \right\rfloor, \quad s = M - m, \quad k = n - M, \quad d = n - 2, \quad J = \left\lfloor \frac{n}{\log n} \right\rfloor.$$

For large n, $s \geq 1$, and associativity gives

$$H \vee K_{n-m} = (H \vee K_s) \vee K_k. \tag{71}$$

Let $X_1, \ldots, X_{n-1}$ be independent unit exponentials and let $T = \min_i X_i$. After deleting the minimizing coordinate, memorylessness gives d independent unit-exponential residuals, independent of T; write their order statistics as $0 < y_1 < \cdots < y_d$. Moreover, $T \sim \mathrm{Exp}(n-1)$.

For the outer join in (71), take cross weight $c_0 = T/n$. A residual y_i assigned to the M-vertex core must be realized there as

$$a_i = y_i + \frac{MT}{n}.$$

Every unselected residual is assigned to K_k as the positive eigenvalue $y_i + kT/n$. It remains to realize $M - 1$ of the a_i on $H \vee K_s$.

Fix an anchor $1 \leq j \leq J$ and put

$$c_1 = \frac{a_j}{M}, \qquad b_j = mc_1 = \frac{my_j}{M} + \frac{mT}{n}, \qquad S_j = y_d - y_j.$$

Use $a_j = Mc_1$ as the distinguished inner-join eigenvalue. Assign the next ℓ residuals and the largest residual to the H block. The resulting small and large H eigenvalues are

$$\mu_i = b_j + y_{j+i} - y_j \quad (1 \leq i \leq \ell), \qquad \mu_{m-1} = b_j + S_j. \tag{72}$$

Choose any further $m - \ell - 2$ indices strictly between $j + \ell$ and d for the middle H eigenvalues and choose $s - 1$ unused indices above j for the K_s block. The count is

$$1 + (m-1) + (s-1) = M - 1,$$

and the choice is possible because $d - j \geq d - J \geq M - 2$ for large n. For an unused index p assigned to the H block, the corresponding value is $\mu = b_j + y_p - y_j$; these values lie strictly between the displayed small and large extremes, so the ordered tail condition applies. If $p > j$ is assigned to K_s, its underlying eigenvalue $a_p - mc_1 = a_p - b_j$ is positive.

Put $q = 1 - (A\ell)^{-1} \in (0, 1)$. Since

$$\sum_{i=1}^{\ell} \mu_i \leq \ell(b_j + S_j - S_{j+\ell}),$$

the tail inequality holds whenever

$$S_{j+\ell} > q(S_j + b_j). \tag{73}$$

Indeed, this is equivalent to $A\ell S_{j+\ell} > (A\ell - 1)(S_j + b_j)$, and hence

$$\mu_{m-1} = S_j + b_j > A\ell(b_j + S_j - S_{j+\ell}) \geq A\sum_{i=1}^{\ell} \mu_i.$$

Thus every good anchor produces the inner realization; two applications of Lemma 10.2 realize the full target multiset.

If all J anchors fail, then

$$S_{j+\ell} \leq q(S_j + b_j) \qquad (1 \leq j \leq J). \tag{74}$$

Let

$$L = \left\lfloor \frac{J-1}{\ell} \right\rfloor + 1, \qquad p = 1 + L\ell.$$

Iteration along $j = 1, 1+\ell, \ldots, 1+(L-1)\ell$ gives

$$S_p \leq q^L S_1 + \frac{q}{1-q} \max_{j \leq J} b_j. \tag{75}$$

Now $p \leq J + \ell$. Since $L \asymp n/(\ell \log n)$ and $-\log q > 0$, one has $q^L \leq 1/n$ for all large n. Also, for all large n,

$$q^L \leq \frac{1}{n}, \qquad \max_{j \leq J} b_j \leq \frac{3m}{n}(y_d + T).$$

Hence, for a constant $C = C(H)$,

$$y_d - y_p = S_p \leq \frac{C}{n}(y_d + T). \tag{76}$$

The number of residuals from y_p through y_d is

$$r = d - p + 1 \geq d - J - \ell + 1 = n - O\left(\frac{n}{\log n}\right).$$

Lemma 11.2 therefore bounds the probability that all moving-anchor charts fail by $\exp\{-n \log n + O_H(n)\}$.

For the lower bound, $H \vee K_{n-m}$ is connected and noncomplete. By continuity, the missing-edge inequality also holds on the relaxed spectral closure, and therefore excludes every positive tuple satisfying $\max_i X_i < n \min_i X_i/(n-2)$. With $d = n - 2$, this event has probability

$$\begin{aligned}(d+1) \int_0^\infty e^{-t} \left(e^{-t} - e^{-(1+2/d)t}\right)^d \, dt \\ = \binom{d(d+3)/2}{d}^{-1} = \exp\{-n \log n + O(n)\}.\end{aligned}$$

This proves (69).

Under strict (A, ℓ)-tail-admissibility the H weighting is strict, both complete blocks are strict, and $c_0, c_1 > 0$ almost surely. Under relaxed inclusion, the construction proves the bound for the relaxed volume defining P; Proposition 2.1, together with the nullity of ties and equality hypersurfaces, shows that no volume estimate changes. □

Corollary 11.4 (Four relaxedly tail-admissible cores)**.** *The following graphs are relaxedly tail-admissible with the indicated parameters:*

H	ℓ	A
P_3	1	3
$K_4 - e$	1	2
Paw	1	$2 + \sqrt{3}$
C_4	2	1

Consequently, for every graph in the table,

$$1 - P(H \vee K_{n-|V(H)|}) = \exp\{-n \log n + O_H(n)\}, \tag{77}$$

or equivalently $1 - P(\mathrm{susp}^k H) = \exp\{-k \log k + O_H(k)\}$.

Proof. The ordered regions contain, respectively,

$$\mu_2 \geq 3\mu_1, \qquad \mu_3 \geq 2\mu_1, \qquad \mu_3 \geq (2+\sqrt{3})\mu_1, \qquad \mu_3 \geq \mu_1 + \mu_2.$$

The strict versions of the displayed inequalities define the stated tail cones and lie in the corresponding relaxed exact regions; in every case $A\ell > 1$. Apply Theorem 11.3 and replace n by $k + |V(H)|$. □

Theorem 11.5 (Optimal law for one and two missing edges)**.** *Let $q \in \{1, 2\}$, and let G_n be obtained from K_n by deleting exactly q edges, with $n \geq 3$ for $q = 1$ and $n \geq 4$ for $q = 2$. Uniformly over the deleted edges,*

$$1 - P(G_n) = \exp\{-n \log n + O(n)\}, \tag{78}$$

or equivalently

$$\lim_{n\to\infty} \frac{-\log(1 - P(G_n))}{n \log n} = 1.$$

For $q = 2$, the two cases are

$$K_n - M_2 = C_4 \vee K_{n-4}, \qquad K_n - E(P_3) = \mathrm{Paw} \vee K_{n-4},$$

according as the deleted edges are disjoint or adjacent.

Proof. For one edge, Corollary 9.2 gives the stronger exact formula (55) and the asymptotic (56). For two edges their union is either M_2 or P_3, giving the displayed decompositions. Corollary 11.4 applies to the two core types C_4 and Paw, so the constants are uniform. This also proves Theorem 1.7. □

12. Four-vertex spectral volume and edge addition

We next record consequences for Question 3 of [5], starting with the complete spectral-volume census for connected four-vertex graphs.

Proposition 12.1 (Four-vertex volume census)**.** *Up to isomorphism, the connected graphs on four vertices have the following spectral volumes:*

G	$P(G)$	*decimal value*
$K_{1,3}$	$\frac{1}{11} + \frac{8\sqrt{33}}{363} \log\left(\frac{161+17\sqrt{33}}{128}\right)$	0.1799703569…
P_4	$\frac{9}{10} + \frac{3\sqrt{5}}{25} \log\left(\frac{3-\sqrt{5}}{3+\sqrt{5}}\right)$	0.3835092708…
Paw	$\frac{9}{13}$	0.692307…
C_4	$\frac{3}{4}$	0.75
$K_4 - e$	$\frac{9}{10}$	0.9
K_4	1	1

Every disconnected graph on four vertices has spectral volume zero.

Proof. Let X_1, X_2, X_3 be independent unit exponentials and write their order statistics as $a \le b \le c$. Put $p = b/a$ and $q = c/a$. For every symmetric homogeneous cone whose ordered ratio region is $\Omega \subseteq \{1 \le p \le q\}$, exponential normalization gives

$$P(G) = 12 \iint_{\Omega} \frac{dp\,dq}{(1+p+q)^3}. \tag{79}$$

Indeed, the factor 6 accounts for permutations, the substitution $(a, b, c) = (a, pa, qa)$ has Jacobian a^2, and $\int_0^\infty a^2 e^{-(1+p+q)a}\,da = 2(1+p+q)^{-3}$.

The exact regions of [5, Theorems 3.1–6.1], after ordering, give

$$\begin{aligned}\Omega_{K_{1,3}} &= \left\{1 \le p < 2 : \frac{2p(3p-1)}{2p-1} \le q \le \frac{2(3-p)}{2-p}\right\} \\ &\quad \cup \left\{p \ge 2 : q \ge \frac{2p(3p-1)}{2p-1}\right\}, \\ \Omega_{P_4} &= \left\{p > 2 : q \ge \frac{p(p-1)}{p-2}\right\}.\end{aligned}$$

Integrating first in q in (79), followed by elementary partial fractions in p, yields the first two displayed values. The ordered regions

$$c \ge (2+\sqrt{3})a, \qquad c \ge a+b, \qquad c \ge 2a$$

give respectively $9/13$, $3/4$, and $9/10$. Complete-graph flexibility gives $P(K_4) = 1$. A disconnected Laplacian has at least two zero eigenvalues, so its normalized region has ambient measure zero. □

For $n \ge 3$, define the connected fixed-order edge gain

$$c_n^{\mathrm{conn}} = \min\left\{P(G+e) - P(G) : \begin{array}{l} G \text{ connected and noncomplete, } |V(G)| = n, \\ e \notin E(G) \end{array}\right\}.$$

Relaxed support makes edge monotonicity immediate: $\mathcal{E}_{\ge 0}(G) \subseteq \mathcal{E}_{\ge 0}(G+e)$, and hence $P(G+e) \ge P(G)$.

Corollary 12.2 (Question 3 in three interpretations)**.** *The following statements hold.*

(i) *If disconnected graphs are admitted literally, Question 3 of* [5] *has a negative answer at every fixed order $n \ge 3$.*

(ii) *No positive edge-gain constant is uniform simultaneously in the graph and its order, even for connected graphs.*

(iii) *In the first two nontrivial connected orders,*

$$c_3^{\mathrm{conn}} = \frac{1}{2}, \qquad c_4^{\mathrm{conn}} = \frac{1}{10}.$$

Proof. For (i), add one edge to the empty graph on $n \ge 3$ vertices. Both graphs remain disconnected and both volumes are zero. For (ii), use $K_n - e \to K_n$. Theorem 11.5 gives the sharper form

$$P(K_n) - P(K_n - e) = \exp\{-n\log n + O(n)\} \longrightarrow 0.$$

For $n = 3$, the only connected noncomplete graph is P_3, with $P(P_3) = 1/2$ and $P(K_3) = 1$.

At order four every edge addition occurs in the list

$$\begin{gathered} K_{1,3} \longrightarrow \text{Paw}, \\ P_4 \longrightarrow \text{Paw} \quad \text{or} \quad P_4 \longrightarrow C_4, \\ \text{Paw} \longrightarrow K_4 - e, \qquad C_4 \longrightarrow K_4 - e, \qquad K_4 - e \longrightarrow K_4. \end{gathered}$$

Comparison in Proposition 12.1 shows that the smallest gain is $1 - 9/10 = 1/10$. □

13. Conclusions and remaining problems

Theorem 3.1 gives an obstruction for $K_n - e$ that depends solely on the extreme eigenvalues. The orbit-first-contact argument proves that this obstruction is exact for the relaxed cone, equivalently for the closure of the strict cone, in every order. The single-contact theorem then shows that the closure loses no positive interior or equality-face point when $n \geq 5$. Together with the explicit $n = 3, 4$ results, this determines the exact strict-support locus in every order. Equality forces the complement of any strict-support graph to be a matching, and Corollary 6.3 shows that the same half-cone is the exact union over all connected noncomplete graphs of fixed order. The exact volume formula and the PSD–SDIEP consequence then follow.

The next exact-support frontier begins with the two defect types $K_n - M_2$ and $K_n - E(P_3)$. The present paper determines their sharp volume defect scale, but not their full semialgebraic spectral cones. For $K_n - M_2$, Theorem 8.2 proves that arbitrary linear complement weights add nothing beyond the symmetric facet (44); hence any full-cone theorem must use nonlinear orbit incidence. Theorem 8.3 completes the first two-dimensional-top-eigenspace slice and isolates the unique order-six endpoint failure.

The remaining directions are now sharply delimited.

(i) For Question 3, the literal version allowing disconnected graphs is false and the all-orders connected version has no positive constant. However, whether $c_n^{\mathrm{conn}} > 0$ for each fixed $n \geq 5$ remains open. The values $c_3^{\mathrm{conn}} = 1/2$ and $c_4^{\mathrm{conn}} = 1/10$ are exact.

(ii) Theorem 10.4 gives exponential convergence for each fixed number of missing edges, while Theorem 11.5 identifies the sharp $n \log n$ exponent for one and two missing edges. It is natural to ask whether the same law holds for every fixed nonzero defect pattern, or more generally to characterize the fixed cores that are tail-admissible. A defect count growing with n lies outside both methods.

(iii) Corollary 9.2 proves strict one-step suspension monotonicity along the one-missing-edge family. Whether $P(\mathrm{susp}\, G) > P(G)$ for every connected noncomplete graph remains open; the iterated limit is settled by Theorem 10.3.

(iv) For $K_n - M_2$, determine whether (44) is sufficient for the relaxed cone, and determine exact strict support in the remaining eigenspace-rank slices, beginning with the unresolved $n = 5$ version of Theorem 8.3. Theorem 8.2 rules out any improvement obtained solely by changing the two coefficients of a linear complement witness. The full cone for the adjacent defect $K_n - E(P_3)$ is also open.

Appendix A. Sign verification for the $K_4 - e$ synthesis

We prove Lemma 6.1. Put

$$R = z - x, \qquad c = \frac{x}{4R}, \qquad \theta = \frac{y - x}{R}.$$

Then $0 < c \leq 1/4$ and $0 \leq \theta \leq 1$. For orthonormal $q, r \in \mathbf{1}^{\perp}$ with $r_1 = 0$ and $q_1 q_2 = c$, the matrix (29) has spectrum $\{0, x, y, z\}$ and

$$L_{12} = -\frac{x}{4} + R q_1 q_2 = 0.$$

For $i \neq j$, write

$$\ell_{ij} = \frac{L_{ij}}{R} = -c + \theta r_i r_j + q_i q_j. \tag{80}$$

Case I: $2x \leq z \leq 5x/2$. Here $1/6 \leq c \leq 1/4$. Define

$$a = \sqrt{c}, \qquad h = \sqrt{\frac{1}{2} - 2c}, \qquad D = 3 - 4c,$$

and set

$$q = (a, a, -a + h, -a - h), \qquad r = \frac{(0, 2h, -2a - h, 2a - h)}{\sqrt{D}}. \tag{81}$$

Since $c \geq 1/6$, one has $0 \leq h \leq a$. Direct calculation gives

$$\mathbf{1}^{\mathsf{T}} q = \mathbf{1}^{\mathsf{T}} r = 0, \qquad \|q\| = \|r\| = 1, \qquad q^{\mathsf{T}} r = 0.$$

Indeed, $\|q\|^2 = 4a^2 + 2h^2 = 1$, and the squared norm of the numerator of r is $8a^2 + 6h^2 = D$. Also $r_1 = 0$ and $q_1 q_2 = a^2 = c$.

The five allowed normalized entries are

$$\begin{aligned}
\ell_{13} &= -c + a(h - a), \\
\ell_{14} &= -c - a(a + h), \\
\ell_{23} &= -c - \theta \frac{2h(2a + h)}{D} + a(h - a), \\
\ell_{24} &= -c + \theta \frac{2h(2a - h)}{D} - a(a + h), \\
\ell_{34} &= -c + \theta \frac{\frac{1}{2} - 6c}{D} + 3c - \frac{1}{2}.
\end{aligned}$$

The first three are strictly negative. For the fourth, $0 \leq \theta \leq 1$ gives

$$\ell_{24} \leq F(c) := -2c - ah + \frac{2h(2a - h)}{D}.$$

Using $h^2 = 1/2 - 2c$,

$$DF(c) = 8c^2 - 2c - 1 + (1 + 4c)ah.$$

Since $ah \leq c$,

$$DF(c) \leq 12c^2 - c - 1 \leq -\frac{1}{2} \qquad (1/6 \leq c \leq 1/4).$$

Thus $\ell_{24} < 0$. Finally, $(1/2 - 6c)/D < 0$, and hence

$$\ell_{34} \leq 2c - \frac{1}{2} \leq 0.$$

Equality is possible only when $c = 1/4$ and $\theta = 0$, which is exactly the ray $z = 2x$, $y = x$.

Case II: $z \geq 5x/2$**.** Here $0 < c \leq 1/6$. Define

$$s = \sqrt{\frac{1}{2} + c}, \qquad d = \sqrt{\frac{1}{2} - 3c}, \qquad b = \frac{s - d}{2}, \qquad A = \frac{s + d}{2} = s - b,$$

and

$$N = 1 + 4c + 4b^2.$$

Set

$$q = (A, b, 0, -s), \qquad r = \frac{(0, s, -s - b, b)}{\sqrt{N}}. \tag{82}$$

One has $A, b, s > 0$ and

$$Ab = \frac{s^2 - d^2}{4} = c.$$

It follows directly that $q, r \in \mathbf{1}^{\perp}$, $q^{\mathsf{T}} r = 0$, and $\|q\| = 1$. Since $c = sb - b^2$, the squared norm of the numerator of r is

$$s^2 + (s + b)^2 + b^2 = 1 + 4c + 4b^2 = N.$$

Thus the required orthonormality and missing-entry identities hold.

Equation (80) now gives

$$\begin{aligned}
\ell_{13} &= -c,\\
\ell_{14} &= -c - As,\\
\ell_{23} &= -c - \theta\frac{s(s+b)}{N},\\
\ell_{24} &= -c - sb + \theta\frac{sb}{N},\\
\ell_{34} &= -c - \theta\frac{b(s+b)}{N}.
\end{aligned}$$

All are strictly negative. For the only mixed-sign expression,

$$\ell_{24} \le -c - sb\left(1 - \frac{1}{N}\right) < 0$$

because $c > 0$ and $N > 1$.

At the seam $z = 5x/2$, the two formulas agree. At $z = 2x$ they reduce to

$$q = \frac{1}{2}(1, 1, -1, -1), \qquad r = \frac{1}{\sqrt{2}}(0, 0, -1, 1),$$

and

$$L_{13} = L_{14} = L_{23} = L_{24} = -\frac{x}{2}, \qquad L_{34} = -\frac{y - x}{2}.$$

This proves Lemma 6.1 and identifies the unique non-strict ray.

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